Design of AI Systems – Module 8 - Group 22

Erik Andreasson (MPDSC) - 200310304498 - eriub@chalmers.se,
William Norland (MPDSC) - 200111085155 - williamnorland@gmail.com

2026-03-22

We the students hereby declare that we have both actively participated in solving every exercise. All solutions are entirely our own work, without having taken part of other solutions.

Hours spent

- Erik Andreasson: 34
- William Norland: 30

Erik Andreasson (MPDSC) - 200310304498 - eriub@chalmers.se,
William Norland (MPDSC) - 200111085155 - williamnorland@gmail.com

Project Introduction

1.1 Background & Topic

Natural Language Processing (NLP) is a subfield of machine learning concerned with enabling computers to understand and work with human language. Despite its central importance to modern AI systems, NLP has not been covered in depth in this course, which was a primary motivation for us to propose this project. Our goal being to use the mini-project as an opportunity to explore this area of AI systems and build intuition for how computers can represent the semantic meaning of words.

A fundamental challenge in NLP is the representation of meaning from textual content. How is the meaning of a word encoded in a form that a machine learning model can work with? The simple approach is one-hot encoding, where each word in a vocabulary of size V is assigned a unique vector in $\mathbb{R}^V$ with a single element set to 1. While simple, this representation has every word equidistant from every other word meaning that the encoding of "king" is no more similar to "queen" than it is to "bicycle". One-hot encoding does not allow for any representation of similarity in meaning or context between words.

Word2Vec, introduced by Mikolov et al. at Google in 2013, addressed this by learning dense, low-dimensional word representations called embeddings directly from raw text. The core insight is the distributional hypothesis being that words that appear in similar contexts tend to have similar meanings. By training a shallow neural network to predict a word from its surrounding context words (or vice versa), the internal weight matrix of the network converges to a representation where semantically related words occupy nearby regions of the embedding space. The result is that geometric relationships in this space encode meaningful structure with the classic example being that the vector arithmetic $\text{king} - \text{man} + \text{woman}$ produces a vector close to queen.

1.2 Objectives

The concrete objectives of this project are:

- Implement the full Word2Vec pipeline from scratch, using PyTorch only for tensor operations. All layers, training logic, and optimization are implemented manually.
- Train and compare two output objectives: full softmax and negative sampling, analyzing both their mathematical formulations and their practical computational trade-offs.
- Study how the volume of training data affects the quality of learned embeddings by training across three corpus sizes.
- Visualize the learned embedding spaces using PCA and UMAP, and evaluate embedding quality qualitatively through nearest-neighbor analysis.

1.3 Relation to Course Concepts

Although Word2Vec originates in NLP, it is related to the latent factor models studied in this course in the context of recommendation systems. In collaborative filtering, the goal is to factorize a sparse user-item interaction matrix $\boldsymbol{R} \approx \boldsymbol{P}\boldsymbol{Q}^T$ into a matrix of latent user factors $\boldsymbol{P}$ and latent item factors $\boldsymbol{Q}$. The core idea being that the latent factors are learned representations that capture the underlying structure of user preferences. This can then be used to make inference on unseen data to for example, recommend movies to users.

Word2Vec can be understood as the same idea applied to language. The interaction matrix is represented via the input embedding matrix $\boldsymbol{W}_{\text{in}}$ and output embedding matrix $\boldsymbol{W}_{\text{out}}$ as the two latent factor matrices. Training pushes words that co-occur frequently into compatible regions of the

latent space similarly to how collaborative filtering pushes users and items with high interaction into compatible latent factor directions.

This project extends beyond simply applying this concept by implementing the full optimization machinery by hand without the use of a packaged library. The forward and backward passes through each layer, the mini-batch gradient descent, and the loss evaluations are all done from scratch. This gives us direct experience in the details of how the embedded representations are actually formed. This enables us to directly understand the impact of certain design choices and hyperparamater values beyond input and output from a completed PyTorch module.

Implementation Methodology

As part of the challenge to our project we decided to implement nearly all aspects of the model from scratch. This includes the full stack of modules and components required to preprocess the data, feed the data to the model, each neural network layer, and train the final result. Our main methodology for the implementation of each component was to focus on creating an inherited structure focused on separation of concerns, reusability, and scalability. Especially for the neural network layers we wanted forward and backward passes to operate over each layer independently of type or internal structure. This would allow us to create multiple models to experiment with.

2.1 Tokenizer

The implementation of our tokenizer converts an input corpus to a vocabulary where words are indexed by a number which represents their tokenized value. These can be used for referencing and dereferencing of words from the training set as well as constructing one-hot encoded vectors. Furthermore, the tokenizer exposes a method for constructing continuous bag of words (CBOW) context and target word pairs for training. The method uses a sliding window over the dataset of sentences, constructing a pair of specified k number of context words along with the missing target word that the model is to predict.

The tokenizer uses a preprocessing method when constructing the vocabulary which allows us to collect as much useful information as possible. Considering that our goal is to analyze the semantic meaning of individual words and not sentences, we made some specific design choices in the preprocessing. Both of our operations were chosen on the basis that our representations do not attempt to represent sentiment, inflection, grammatical usage, or any other contextualized differences between word representations. Instead, we focus on representing and comparing the words in isolation. Therefore, we perform the following operations:

- Remove punctuation: The inclusion of punctuation would be misleading since it is a tool for expressing meaning in the context of a sentence. For instance a question, a pause, an exclamation, and so on. They are grammatical tools which do not alter the meaning of the word they are attached to. For instance, “apple!”, “apple?”, “apple,”, and “apple.” all have the word apple which we aim to represent, not the inflection of the word.
- Convert to lowercase: This keeps all words equal before their representation, removing possible differences in representation for capitalized words such as “Apple” and “apple”.

2.2 Dataloader

The dataloader is responsible for feeding batched data to the model during each epoch. Two implementations were developed to address different memory and performance constraints.

The base DataLoader class operates over a pre-allocated list of all training context-target pairs. It supports optional shuffling via index permutation and yields batches of a fixed size by slicing this

list on each iteration. While straightforward, this approach requires all context-target pairs to be held in memory simultaneously which becomes impractical as corpus size grows, since the number of CBOW pairs scales with both the number of sentences and the window size. Having a maximum number of sentences larger than 200 while one-hot encoding was enabled as output often crashed our computers from running out of memory.

To address this, StreamingDataLoader was introduced. Rather than storing all pairs upfront, it accepts a generator function that produces context-target pairs on demand. The total pair count is determined at construction time by exhausting the generator once without retaining any pairs, keeping peak memory at the size of a single sample regardless of corpus size. Pairs are accumulated into batches incrementally and yielded as they fill, with any remaining pairs at the end of a dataset sweep yielded as a final partial batch.

The downside is that StreamingDataLoader does not support shuffling since pairs are always yielded in corpus order with no index structure to permute. For the purposes of this project this was deemed acceptable, since the sliding window provides variation in the input and ordered context-target pairs should not considerably affect the final result.

All training runs in this project use StreamingDataLoader, with a separate generator call for the training split (sentences 0–80%) and validation split (80–100%), controlled by the start and end fraction parameters exposed by the tokenizer's generate_CBOW_pairs method.

2.3 Layers

We chose to define our layers in an object oriented way. The reason for this being that we aim to increase the readability of our code by relating python objects to mathematical objects. Hopefully, creating a successful bridge between the theory and the implementation.

Each layer of our model can be modeled as a function.

$$\hat{\boldsymbol{y}} = f(\boldsymbol{v})$$

This function can contain any arbitrary operation on the input vector given that it returns a Tensor object as a result. This effectively functions as an interface for clients to use.

2.3.a Softmax Activation

Central to the straightforward implementation of the model is the softmax activation function which allows the final linear layer to output a vector of predicted probabilities for the target word across the entire tokenizer vocabulary. This is done by turning a vector of numbers $\boldsymbol{z} \in \boldsymbol{R}^n$ in to a vector of probabilities $\hat{\boldsymbol{y}}$ s.t $\sum_i \hat{y}_i = 1$. The softmax function is defined as:

$$\hat{\boldsymbol{y}}_i = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$$

The gradients are calculated automatically during the backward pass and are set as follows:

$$\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

$$\frac{\partial \hat{y}_i}{\partial z_j} = \hat{y}_i(\delta_{ij} - \hat{y}_j)$$

$$\text{diag}(\hat{\boldsymbol{y}}) = \begin{pmatrix} \hat{y}_1 & 0 & \dots \\ 0 & \hat{y}_2 & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T = \begin{pmatrix} \hat{y}_1^2 & \hat{y}_1\hat{y}_2 & \dots \\ \hat{y}_2\hat{y}_1 & \hat{y}_2^2 & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}} = \text{diag}(\hat{\boldsymbol{y}}) - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T$$

The single element version $\frac{\partial \hat{y}_i}{\partial z_j}$ of the derivative is included for clarity as it makes it easier to understand the full vector ($\hat{\boldsymbol{y}}$).

In order to calculate the partial derivative of the loss given the input $\boldsymbol{z}$ to the activation layer we can apply the upstream partial derivative denoted $\frac{\partial L}{\partial \hat{\boldsymbol{y}}}$ using the chain rule:

$$\frac{\partial L}{\partial \boldsymbol{z}} = \left(\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}}\right)^T \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{z}} = (\text{diag}(\hat{\boldsymbol{y}}) - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T) \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{z}} = \text{diag}(\hat{\boldsymbol{y}}) \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}} - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{z}} = \hat{\boldsymbol{y}} \odot \frac{\partial L}{\partial \hat{\boldsymbol{y}}} - \left(\hat{\boldsymbol{y}}^T \frac{\partial L}{\partial \hat{\boldsymbol{y}}}\right)\hat{\boldsymbol{y}}$$

Where the first transposition is required to match the dimensionalities since we are working with column vectors. Furthermore, the last line follows from simplifications of the diagonal matrix multiplied by a vector reducing to an element-wise scale of the upstream partial derivative and the second line results from the dot product reducing the vector multiplication to a scalar multiplication. Note that $\left(\text{diag}(\hat{\boldsymbol{y}}) - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T\right)^T = \text{diag}(\hat{\boldsymbol{y}}) - \hat{\boldsymbol{y}}\hat{\boldsymbol{y}}^T$ due to symmetry.

2.3.b Linear Layer

The object `LinearLayer2D` performs the following operation on an input vector $\boldsymbol{v}$:

$$\boldsymbol{z} = \boldsymbol{v}^T\boldsymbol{W} + \boldsymbol{b}$$
$$\hat{\boldsymbol{y}} = \boldsymbol{\sigma}(\boldsymbol{z})$$

Where $\boldsymbol{b}$ is the bias term, $\boldsymbol{W}$ is the weights, and $\boldsymbol{\sigma}$ is the activation function, $\boldsymbol{z}$ is just an intermediary step used for convenience.

We have, however, modified this operation slightly so that we can remove the separate bias term and include it directly in an expanded weight matrix $\boldsymbol{W_b}$. This is achieved by adding a row filled with ones to $\boldsymbol{W}$. So the operation actually becomes:

$$\boldsymbol{z} = \boldsymbol{v}^T\boldsymbol{W_b}$$
$$\hat{\boldsymbol{y}} = \boldsymbol{\sigma}(\boldsymbol{z})$$

The gradients are calculated as following during the backward pass:

$\hat{\boldsymbol{y}}$ w.r.t the input vector $\boldsymbol{v}$

$$z_i = v_1 \boldsymbol{W_b}_{1i} + v_2 \boldsymbol{W_b}_{2i} + ... + v_i \boldsymbol{W_b}_{ii} + ... + v_N \boldsymbol{W_b}_{ni}$$

$$\frac{\partial \boldsymbol{z_i}}{\partial \boldsymbol{v_j}} = \boldsymbol{W_b}_{ji}$$

$$\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{v}} = \begin{pmatrix} \frac{\partial z_1}{\partial v_1} & \frac{\partial z_1}{\partial v_2} & \cdots & \frac{\partial z_1}{\partial v_N} \\ \frac{\partial z_2}{\partial v_1} & \frac{\partial z_2}{\partial v_2} & \cdots & \frac{\partial z_2}{\partial v_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial z_M}{\partial v_1} & \frac{\partial z_M}{\partial v_2} & \cdots & \frac{\partial z_M}{\partial v_N} \end{pmatrix} = \boldsymbol{W_b}^T$$

$$\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}} \text{ is taken from the activation function with } \frac{\partial L}{\partial \hat{\boldsymbol{y}}} \text{ applied}$$

$$\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{v}} = \frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}} \cdot \boldsymbol{W_b}^T$$

Therefore, we can calculate $\frac{\partial L}{\partial \boldsymbol{v}}$ with:

$$\frac{\partial L}{\partial \boldsymbol{v}} = \left(\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{v}}\right)^T \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{v}} = \left(\left(\boldsymbol{W_b}^T\right)^T \cdot \left(\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}}\right)^T\right) \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{v}} = \boldsymbol{W_b} \cdot \left(\frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z}}\right)^T \cdot \frac{\partial L}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial L}{\partial \boldsymbol{v}} = \boldsymbol{W_b} \cdot \frac{\partial L}{\partial \boldsymbol{z}}$$

$\hat{\boldsymbol{y}}$ w.r.t the weight matrix $\boldsymbol{W_b}$

$$z_i = v_1 \boldsymbol{W_b}_{1i} + v_2 \boldsymbol{W_b}_{2i} + ... + v_i \boldsymbol{W_b}_{ii} + ... + v_N \boldsymbol{W_b}_{ni} + 1 \cdot \boldsymbol{W_b}_{(n+1)i}$$

$$\frac{\partial z_i}{\partial \boldsymbol{W_b}_{kj}} = \begin{cases} v_k \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

$$\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{W_b}} = \left[\begin{pmatrix} v_1 & 0 & ... & 0 \\ v_2 & 0 & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ v_N & 0 & ... & 0 \\ 1 & 0 & ... & 0 \end{pmatrix} \middle| \begin{pmatrix} 0 & v_1 & ... & 0 \\ 0 & v_2 & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & v_N & ... & 0 \\ 0 & 1 & ... & 0 \end{pmatrix} \middle| ... \middle| \begin{pmatrix} 0 & 0 & ... & v_1 \\ 0 & 0 & ... & v_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & ... & v_N \\ 0 & 0 & ... & 1 \end{pmatrix} \right]$$

Therefore, we can calculate $\frac{\partial L}{\partial \boldsymbol{W_b}}$ where $\frac{\partial L}{\partial \boldsymbol{z}}$ is our upstream gradient. We can do this by applying the chain rule element-wise:

$$\frac{\partial L}{\partial \boldsymbol{W_b}_{ij}} = \frac{\partial L}{\partial z_j} \cdot \frac{\partial z_j}{\partial \boldsymbol{W_b}_{ij}} = \frac{\partial L}{\partial z_j} \cdot v_{b,i}$$

This is equivalent to the outer product:

$$\frac{\partial L}{\partial \boldsymbol{W_b}} = \boldsymbol{v_b}\left(\frac{\partial L}{\partial \boldsymbol{z}}\right)^T$$

$$\frac{\partial L}{\partial \boldsymbol{W_b}} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \\ 1 \end{pmatrix} \begin{pmatrix} \frac{\partial L}{\partial z_1} & \frac{\partial L}{\partial z_2} & \cdots & \frac{\partial L}{\partial z_M} \end{pmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{W_b}} = \begin{pmatrix} v_1\frac{\partial L}{\partial z_1} & v_1\frac{\partial L}{\partial z_2} & \cdots & v_1\frac{\partial L}{\partial z_M} \\ v_2\frac{\partial L}{\partial z_1} & v_2\frac{\partial L}{\partial z_2} & \cdots & v_2\frac{\partial L}{\partial z_M} \\ \vdots & \vdots & \ddots & \vdots \\ v_N\frac{\partial L}{\partial z_1} & v_N\frac{\partial L}{\partial z_2} & \cdots & v_N\frac{\partial L}{\partial z_M} \\ \frac{\partial L}{\partial z_1} & \frac{\partial L}{\partial z_2} & \cdots & \frac{\partial L}{\partial z_M} \end{pmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{W_b}} = \boldsymbol{v_b} \otimes \frac{\partial L}{\partial \boldsymbol{z}}$$

2.3.c Averaging Layer:

The `LinearLayerAvgMixer` averages context embeddings by:

1. Encoding each context word index as a one-hot vector $e_i \in \mathbb{R}^V$
2. Multiplying by the weight matrix: $e_i W = W_{[i]}$ (row i of W)
3. Averaging the results

Step 1 and 2 together are equivalent to a direct row lookup, since multiplying a one-hot vector by a matrix simply selects the corresponding row:

$$\frac{1}{n}\sum_{k=1}^{n} e_{i_k} W = \frac{1}{n}\sum_{k=1}^{n} W_{[i_k]}$$

`EmbeddingAverageLayer` implements this directly without large matrix multiplication and memory costly allocations. It is a mathematically equivalent simplification used to speed up training.

2.4 Loss

2.4.a Binary Cross Entropy Loss

The Binary cross entropy loss function is a loss function that is primarily used to compare probability vectors (vectors where elements sum to 1) against binary vectors (where a single element is set to 1 and there rest are set to 0.) This is usually used in binary classification where a model predicts a probability of a sample having some label, for example, cat, dog, fish. In binary classification there is only a single correct label but to improve model performance we allow it to be kind of uncertain, so it can predict 90% dog, 5% cat, 5% fish.

The loss is given by:

$$L = -y\log(\hat{y}) + (1-y)\log(1-\hat{y})$$

2.4.b Negative Sampling Loss

Instead of computing softmax over the full vocabulary, negative sampling reformulates the problem as binary classification. For a context vector $\boldsymbol{h}$ and target word index p, the loss is:

$$L = -\log \sigma(\boldsymbol{h} \cdot \boldsymbol{w}_p) - \sum_{i=1}^{k} \log\left(1 - \sigma\left(\boldsymbol{h} \cdot \boldsymbol{w}_{n_i}\right)\right)$$

Where $\boldsymbol{w}_p$ is the output embedding of the positive (target) word, $\boldsymbol{w}_{n_i}$ are the output embeddings of k randomly sampled negative words, and σ is the sigmoid function.

The gradients are:

$$\frac{\partial L}{\partial \boldsymbol{w}_p} = (\sigma(\boldsymbol{h} \cdot \boldsymbol{w}_p) - 1) \cdot \boldsymbol{h}$$

$$\frac{\partial L}{\partial \boldsymbol{w}_{n_i}} = \sigma\left(\boldsymbol{h} \cdot \boldsymbol{w}_{n_i}\right) \cdot \boldsymbol{h}$$

$$\frac{\partial L}{\partial \boldsymbol{h}} = (\sigma(\boldsymbol{h} \cdot \boldsymbol{w}_p) - 1) \cdot \boldsymbol{w}_p + \sum_{i=1}^{k} \sigma\left(\boldsymbol{h} \cdot \boldsymbol{w}_{n_i}\right) \cdot \boldsymbol{w}_{n_i}$$

Negatives are sampled from a smoothed unigram distribution $P(w) \propto f(w)^{0.75}$ where $f(w)$ is the word frequency. The smoothing reduces the dominance of very frequent words.

This loss method was implemented to improve the efficiency of training and was discussed by Google in their implementation.

2.5 Training

Training is performed using mini-batch stochastic gradient descent (SGD). Each epoch iterates over the full training set in batches of 32 context-target pairs. Within each batch, gradients are accumulated sample-by-sample and then applied once as an averaged update at the end of the batch.

Two separate training loops were implemented: train for the softmax CBOW model and train_ns for the negative sampling models. The reason for this split is that the NegativeSamplingLoss object maintains its own trainable output embeddings which sit outside the Model object. This means its weight update cannot be handled by the model's backpropagate method and must instead be called explicitly inside the training loop. Combining these into a single universal training loop is a part of potential future work on the project.

Both loops track training and validation loss per epoch. The validation pass runs forward passes over the held-out 20% of sentences without accumulating or applying any gradients, using clear_caches to discard the cached forward-pass inputs that would otherwise accumulate unused.

The final optimized models were trained with the following hyperparameters, kept fixed across all three corpus sizes so that only data volume varied between them:

Hyperparameter	**Value**
Embedding dimension	50
Negative samples k	5
Learning rate	0.2
Batch size	32
Epochs	10
Train / val split	80% / 20%

Table 1: Final hyperparameters for model training and evaluation

The learning rate of 0.2 is relatively high compared to typical neural network deep learning settings, but is appropriate here since the model is shallow and the gradients are sparse resulting in each

update only touching the embedding rows corresponding to the context words and sampled negatives as opposed to the usual full weight matrix.

Model Evaluation

3.1 Visualization methods

3.1.a PCA:
PCA is one of the oldest tricks in the book when it comes to reducing dimensionality of data in order to make it possible to visualize in a 2d, and sometimes 3d, space. Its foundational paper was released all the way back in 1933 by a Harald Hotelling, making it almost 100 years old.

PCA relies on two important concepts from the field of linear algebra, eigenvectors and matrix transformations (base changes to be more exact). The PCA method works in several steps:

- Initially the eigenvectors of the data's covariance matrix are computed. Each eigenvector $\boldsymbol{\lambda}_i$ defines a direction in the feature space, and its corresponding eigenvalue measures how much of the total variance in the data is explained by that direction. For a dataset with three features the covariance matrix and its eigenvectors look like:

$$\begin{pmatrix} 1 & \sigma_{12}^2 & \sigma_{13}^2 \\ \sigma_{12}^2 & 1 & \sigma_{23}^2 \\ \sigma_{13}^2 & \sigma_{23}^2 & 1 \end{pmatrix} \rightarrow \boldsymbol{\lambda}_1 = \begin{pmatrix} \lambda_{11} \\ \lambda_{21} \\ \lambda_{31} \end{pmatrix}, \boldsymbol{\lambda}_2 = \begin{pmatrix} \lambda_{12} \\ \lambda_{22} \\ \lambda_{32} \end{pmatrix}, \boldsymbol{\lambda}_3 = \begin{pmatrix} \lambda_{13} \\ \lambda_{23} \\ \lambda_{33} \end{pmatrix}$$

- The eigenvectors are then sorted by their eigenvalues in descending order, so that the first principal component $\boldsymbol{\lambda}_1$ captures the direction of greatest variance, the second the next greatest, and so on.
- Finally, the top k eigenvectors are selected and used to form a projection matrix $\boldsymbol{P} \in \mathbb{R}^{n \times k}$. The original data is projected into this lower-dimensional subspace via $\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{P}$, producing the reduced representation used for visualisation.

A key limitation of PCA is that it is a purely linear method, meaning it can only capture structure that lies along straight axes in the original space. Non-linear relationships between words, such as semantic clusters that form curved or folded surfaces in embedding space, are invisible to PCA and may be better revealed by UMAP below.

3.1.b UMAP:
UMAP (Uniform Manifold Approximation and Projection) is a considerably more recent method, introduced by McInnes, Healy and Melville in 2018. Unlike PCA, which is a linear technique restricted to capturing variance along orthogonal axes, UMAP is a non-linear dimensionality reduction method built on the theory of Riemannian geometry and algebraic topology.

The core idea is that high-dimensional data is assumed to lie on a low-dimensional manifold embedded in the ambient space. UMAP constructs a weighted graph representing the local neighborhood structure of the data, where the weight of the edge between points x_i and x_j is given by a fuzzy membership strength:

$$\overline{p}_{ij} = p_{ij} + p_{ji} - p_{ij} \cdot p_{ji}, \quad p_{ij} = \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right)$$

where ρ_i is the distance from x_i to its nearest neighbor and σ_i is a per-point bandwidth chosen so that the total weight out of x_i equals $\log_2(k)$ for the chosen number of neighbors k. A

corresponding low-dimensional graph is then optimised to have a similar fuzzy structure, minimising the cross-entropy between the two representations:

$$\mathcal{L} = -\sum_{i \neq j} \left[\overline{p}_{ij} \log q_{ij} + \left(1 - \overline{p}_{ij}\right) \log\left(1 - q_{ij}\right) \right]$$

Two hyperparameters control the resulting layout. `n_neighbors` affects how much local versus global structure is preserved where small values emphasise tight local clusters while larger values give a broader picture of the overall topology of the data. `min_dist` controls how tightly points can be packed together in the embedding; a smaller value produces more compact clusters at the cost of less faithful representation of inter-cluster distances. The cosine metric is used here rather than Euclidean distance, which is more appropriate for word embedding vectors where the direction of the vector encodes meaning rather than its magnitude.

3.2 Results

3.2.a Training Curves

The training and validation loss curves across all three corpus sizes are shown in Figure 1. All three models show consistent decreasing loss across epochs, however, they differ in levels of overfitting where the small corpus model shows signs of potential future improvement. Furthermore, the medium corpus size begins overfitting after epoch 4 while the large corpus model continues to slowly improve with consistent generalization gap. These differences likely stem from the vocabulary size affecting how diluted the gradient update becomes for each sample of a context-target pair. Higher vocabulary size meaning a single sample causes less of an impact on the total gradient for the batch. Nevertheless, as expected, the larger corpus models converge to lower absolute loss values, reflecting that more training data provides richer co-occurrence for the embeddings to learn from.

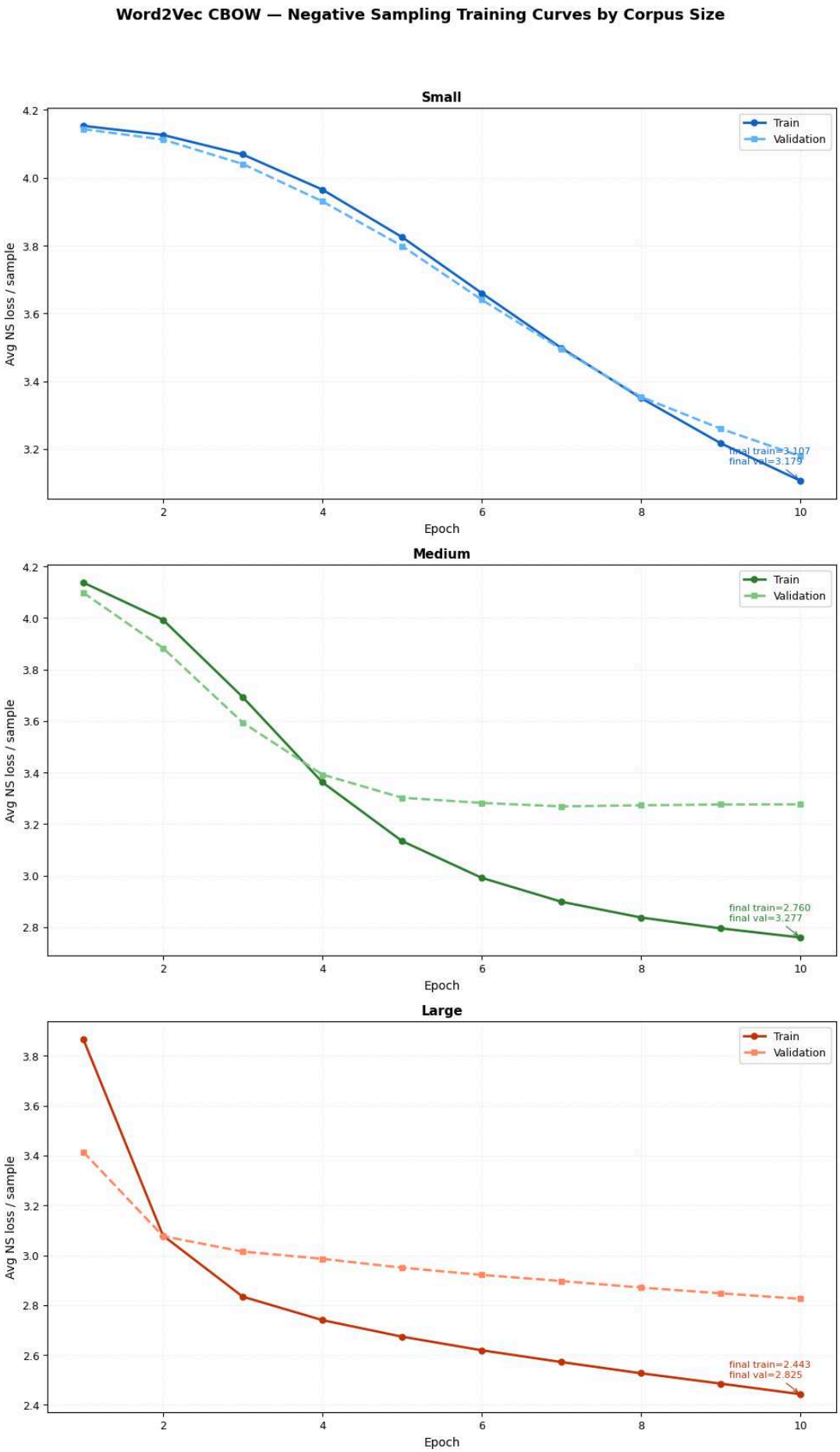


Figure 1: Negative sampling training and validation loss per epoch across Small, Medium, and Large corpus sizes.

3.2.b Nearest Neighbor Analysis

To evaluate the quality of the learned embeddings qualitatively, we query the k nearest neighbors of a set of probe words by cosine similarity. The results for the large corpus model are shown in Table 2, and a cross-scale comparison for a selection of probe words is shown in Table 3.

The cross-scale comparison reveals a clear progression in embedding quality as corpus size increases. On the small corpus (200 sentences), the vocabulary is severely limited with for example "king" being absent entirely. The nearest neighbors seem to be represented by noise, dominated by stop words and random co-occurrences with no discernible semantic pattern. On the medium corpus (1000 sentences), cosine similarities are higher but neighbors are still predominantly function words such as "in", "for", "and", reflecting that the model has learned syntactic context more than semantic meaning. On the large corpus (5000 sentences), the results improve considerably. For example, the nearest neighbors of "first" include "second", "best", and "new" which are words with directly similar meaning where "second" naturally comes after "first" and "best" as well as "new" showing learned understanding of the meaning of primality and uniqueness of being first. Similarly, "world" retrieves words like "game", "league", and "american", which is plausible to frequently co-occur in a Wikipedia-style text of the source corpus. Here it is clear that our choice to not implement a shuffle in the dataloader may be severely affecting the results where articles using words in specific contexts can heavily skew the results.

While it has learned some plausible representation of words, the large corpus results still fall short of what a production Word2Vec model would produce. The neighbors of "king" include "project", "season", and "rock" which are clearly semantically unrelated words. Furthermore, "war" retrieves "season" as its closest neighbor, which may be an indication of co-occurence in topics such as sports in the corpus rather than genuine semantic understanding. The high baseline cosine similarities across all queries (most neighbors scoring above 0.85) suggest the embedding space has not fully separated into distinct semantic regions, with many words still clustered too closely together. This is consistent with the relatively small corpus size and limited training epochs compared to the original Word2Vec paper, which trained on a corpus of roughly 100 billion words.

Query	#1	#2	#3	#4	#5	#6	#7	#8
time	british (0.906)	day (0.893)	same (0.887)	team (0.883)	first (0.881)	end (0.870)	north (0.870)	new (0.867)
year	only (0.885)	but (0.877)	time (0.856)	new (0.853)	egyptian (0.853)	british (0.850)	made (0.848)	half (0.847)
century	following (0.860)	before (0.854)	under (0.845)	at (0.840)	during (0.829)	against (0.828)	from (0.825)	team (0.823)
age	archaeology (0.741)	best (0.728)	ga (0.727)	number (0.722)	four (0.721)	university (0.721)	out (0.717)	townsend (0.716)
world	game (0.872)	british (0.853)	second (0.850)	early (0.833)	league (0.830)	american (0.827)	top (0.825)	school (0.822)
city	first (0.888)	top (0.877)	team (0.864)	end (0.857)	second (0.851)	british (0.851)	site (0.851)	season (0.848)
war	season (0.870)	new (0.866)	british (0.863)	first (0.860)	all (0.849)	end (0.849)	team (0.849)	state (0.846)
government	team (0.856)	gods (0.839)	song (0.836)	army (0.834)	site (0.830)	film (0.826)	fleet (0.826)	album (0.825)
american	game (0.896)	season (0.878)	league (0.875)	only (0.867)	british (0.867)	national (0.865)	three (0.863)	north (0.860)
people	an (0.876)	out (0.871)	many (0.865)	some (0.859)	their (0.856)	these (0.854)	each (0.851)	all (0.851)
king	project (0.827)	season (0.753)	team (0.753)	rock (0.752)	building (0.750)	area (0.750)	although (0.747)	before (0.744)
queen	player (0.641)	aviation (0.576)	rays (0.562)	as (0.560)	for (0.553)	full (0.549)	association (0.546)	or (0.541)
church	film (0.856)	album (0.826)	ship (0.819)	second (0.817)	project (0.811)	team (0.807)	song (0.807)	last (0.806)
new	second (0.912)	first (0.908)	season (0.893)	british (0.891)	only (0.890)	best (0.887)	two (0.885)	all (0.883)
great	side (0.784)	third (0.777)	made (0.773)	main (0.767)	club (0.763)	season (0.761)	album (0.758)	school (0.755)
first	second (0.936)	season (0.917)	new (0.908)	team (0.904)	best (0.903)	film (0.896)	york (0.895)	all (0.894)
history	some (0.879)	first (0.877)	games (0.868)	north (0.865)	its (0.863)	all (0.861)	their (0.857)	new (0.852)

Table 2: K-Nearest neighbor results for probe words on large model with $k = 8$

Query	Model	Top-5 neighbors (Cosine Similarity)
time	Small	mary (0.398), available (0.391), over (0.371), major (0.354), removed (0.353)
	Medium	in (0.817), at (0.791), and (0.782), was (0.782), <PAD> (0.780)
	Large	british (0.906), day (0.893), same (0.887), team (0.883), first (0.881)
world	Small	waldrons (0.415), 1911 (0.410), has (0.404), such (0.399), form (0.395)
	Medium	gods (0.749), game (0.748), of (0.746), while (0.738), for (0.735)
	Large	game (0.872), british (0.853), second (0.850), early (0.833), league (0.830)
war	Small	the (0.510), were (0.503), and (0.497), was (0.493), rock (0.473)
	Medium	between (0.643), only (0.625), may (0.624), was (0.615), with (0.609)
	Large	season (0.870), new (0.866), british (0.863), first (0.860), all (0.849)
king	Small	*word not in vocabulary*
	Medium	blackie (0.525), during (0.522), they (0.498), to (0.496), has (0.488)
	Large	project (0.827), season (0.753), team (0.753), rock (0.752), building (0.750)
first	Small	once (0.488), by (0.447), unk (0.432), that (0.431), an (0.422)
	Medium	for (0.859), with (0.833), and (0.832), is (0.831), in (0.828)
	Large	second (0.936), season (0.917), new (0.908), team (0.904), best (0.903)

Table 3: K-Nearest neighbor results for probe words on all three model with $k = 5$

3.2.c PCA Visualization

Figure 2 shows a PCA projection of the 150 most frequent words from the large corpus embedding. Points are colored by log word frequency. The projection captures only a modest fraction of the total

variance across the 50-dimensional embedding space, which is an expected result. It follows from PCA's linear projection not being able to fully represent the non-linear structure of a learned embedding space in only two dimensions.

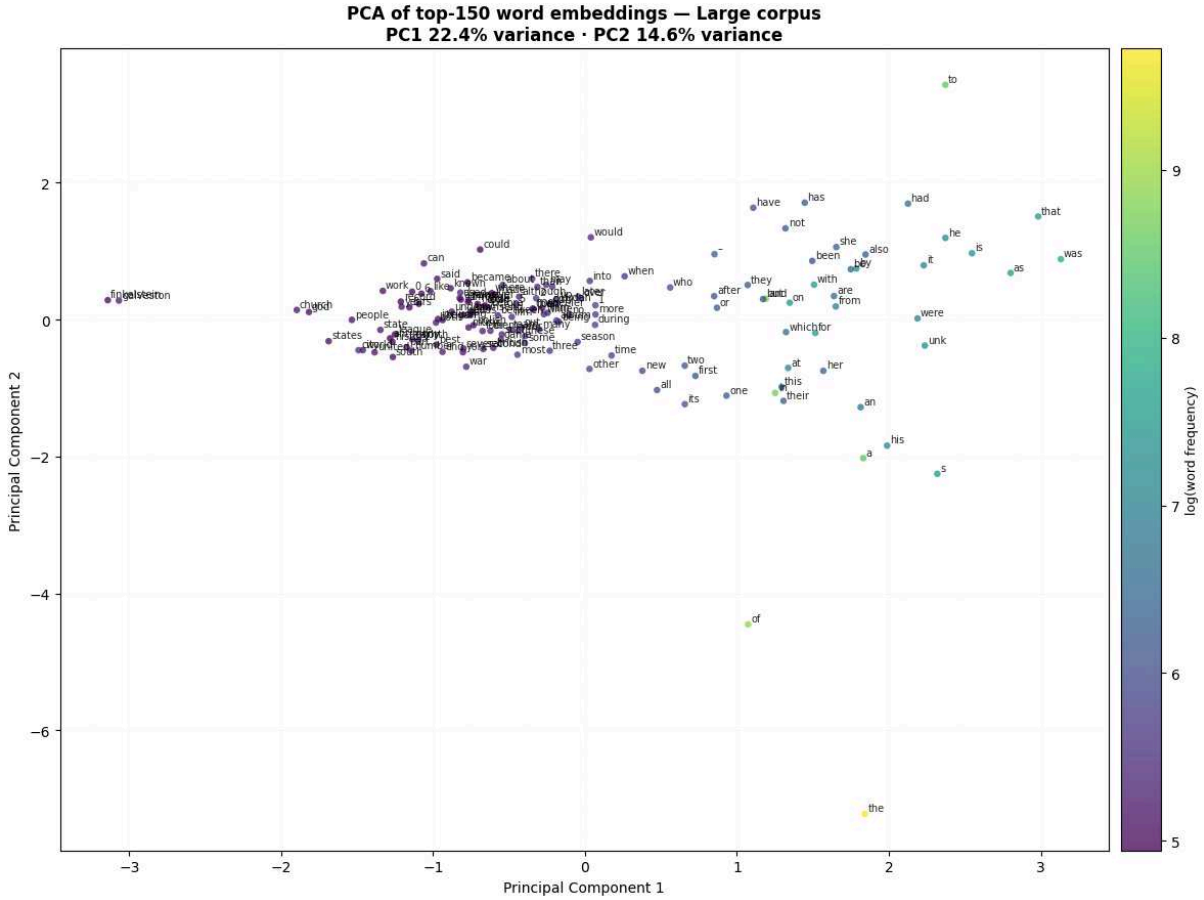


Figure 2: PCA projection of the 150 most frequent word embeddings from the large corpus model. Color indicates log word frequency.

3.2.d UMAP Visualization

Figure 3 shows the same embeddings projected using UMAP with cosine distance, $k = 15$ neighbors and min_dist $= 0.1$. UMAP is better suited to this task than PCA since it preserves local neighborhood structure and can represent non-linear relationships in the embedding space. This superiority over PCA for word embedding representations is clear even in our extremely limited corpus where we observe that numerical words of numbers are tightly clustered in their own space away from the other words. However words like "four" and "number" are not recognized as being numerical in the same way. Nevertheless, local word clusters exist that make sense. Such as "team", "world", "game", and "league" occurring in a cluster at approximately (4.5, −1)

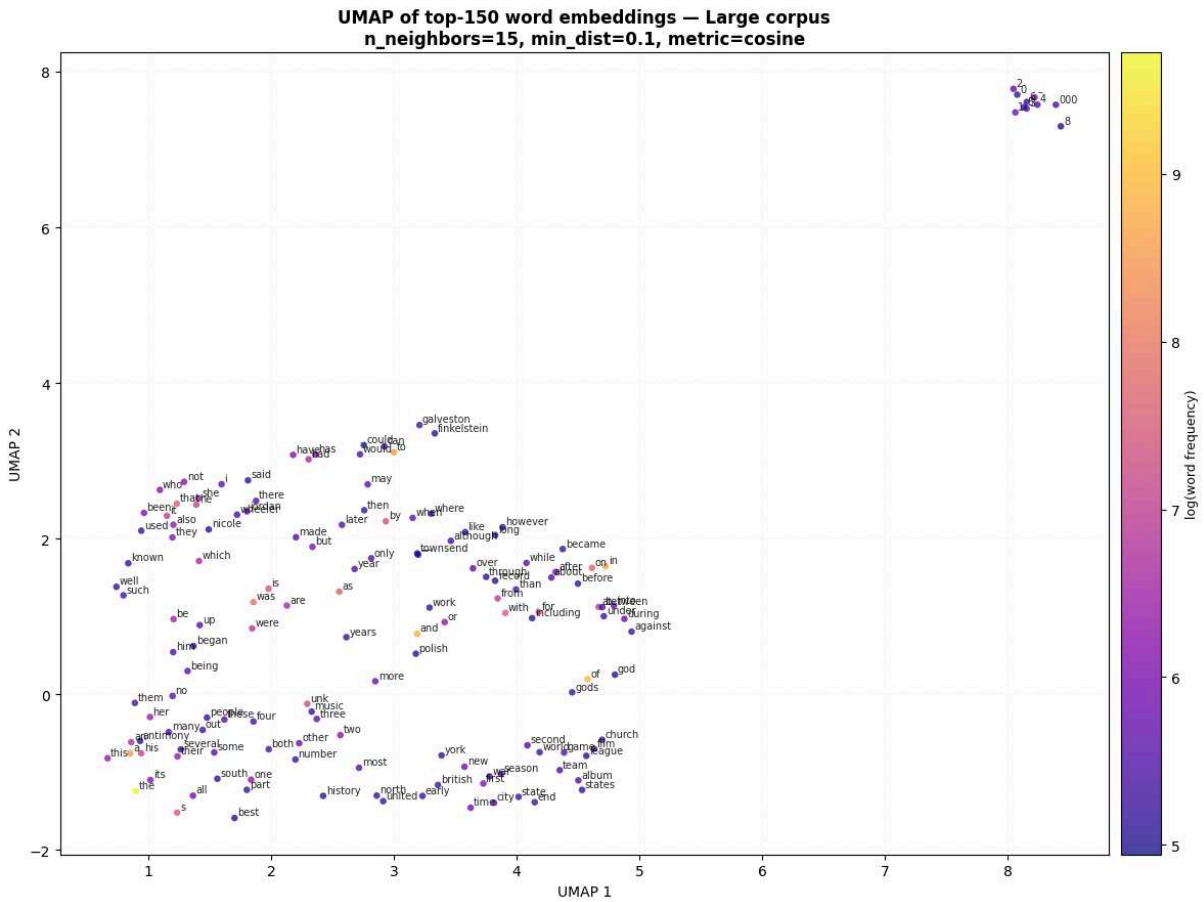


Figure 3: UMAP projection of the 150 most frequent word embeddings from the large corpus model using cosine distance.

Compared to the PCA plot, the UMAP projection produces slightly more separated local clusters, suggesting that the embeddings do contain local neighborhood structure that PCA flattens out. However, the overall layout remains relatively diffuse without any sharply defined semantic groupings besides the number words. This is consistent with our previous observations from k-nearest neighbors of a model that has learned partial but not complete semantic structure given the corpus size.

Finally a PCA comparison plot for corpus sizes is given in Figure 4.

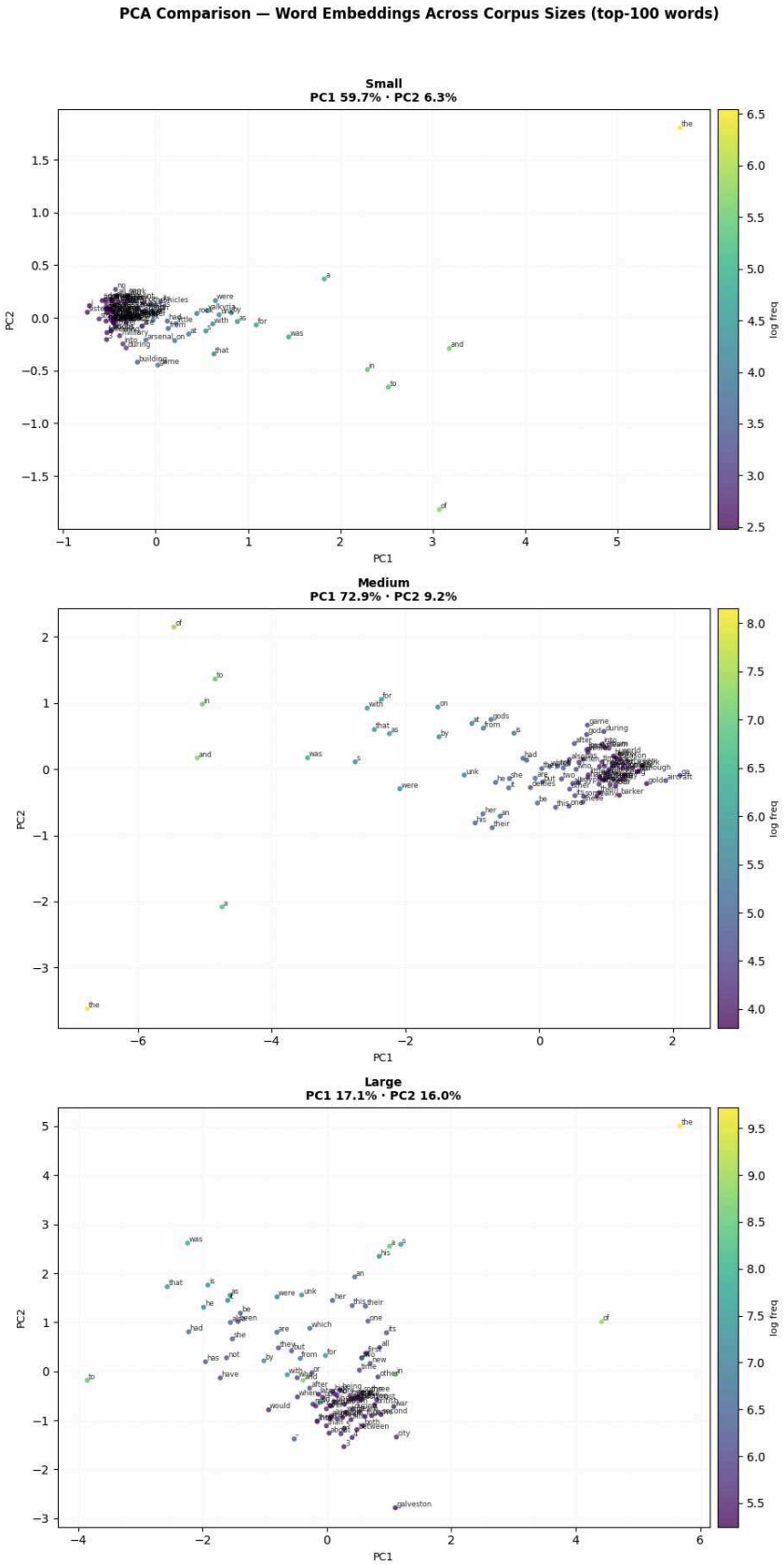


Figure 4: PCA projection of the 100 most frequent word embeddings from each corpus model. Color indicates log word frequency.

Conclusion

This project set out to replicate the core Word2Vec pipeline from scratch using only PyTorch for tensor operations, and to study how corpus size and output objective affect the quality of the learned embeddings.

The implementation successfully reproduces the key components described in the original 2013 paper which includes a tokenizer with preprocessing, a CBOW context-target pair generator, a manually implemented embedding layer with mini-batch gradient descent, and a negative sampling loss with smoothed unigram noise distribution. Three model variants were compared: softmax CBOW, unoptimized negative sampling CBOW, and an optimized embedding lookup model which reveals an interesting practical finding. Theoretical algorithmic complexity does not always translate directly to empirical runtime. The unoptimized negative sampling model was slower than the softmax model at small vocabulary sizes, because the Python loop overhead of individual sigmoid operations outweighed the savings from avoiding the full softmax normalization. The optimized `EmbeddingAverageLayer`, which replaces the dense one-hot matrix multiply with a direct row lookup, was required to achieve the required speedup at scale.

The nearest neighbor analysis and visualizations confirm that the large corpus model has learned meaningful embedding structure to some degree. Words like “first” and “second” cluster together, and contextually co-occurring words appear as neighbors with high cosine similarity. The UMAP visualization further revealed that numerical words form a tight isolated cluster in the embedding space, demonstrating that the model has identified at least one coherent semantic category without any explicit supervision. However, the results also make clear that 5000 sentences is far from sufficient to produce the kind of clean semantic geometry that the original Word2Vec model demonstrated. The embedding space has learned co-occurrence patterns but has not yet fully separated into distinct semantic regions, evidenced by the uniformly high cosine similarities across probe words and semantically unrelated nearest neighbors for words like “king” and “war”. The lack of shuffling in the dataloader is likely a contributing factor, as the sequential ordering of Wikipedia articles means that topic-specific co-occurrence patterns can dominate the learned representations.

The primary limiting factor is corpus size. The original Word2Vec was trained on approximately 100 billion words while our largest model used roughly 5000 sentences from WikiText. Scaling further is the most direct path to better embeddings, but doing so on CPU with a pure Python training loop is prohibitively slow. GPU parallelization of the embedding lookup and gradient accumulation is therefore the most important direction for future work. Implementing shuffling in the streaming dataloader and extending the implementation to support Skip-gram in addition to CBOW, which was planned in the original proposal but not completed due to time constraints, are also natural next steps.

In relation to the course, this project demonstrates concretely how a latent factor model extends beyond recommendation systems into the domain of language. The embedding matrices play the same conceptual role as the latent user and item factor matrices in collaborative filtering. They are both learned representations that compress a high-dimensional co-occurrence structure into a dense low-dimensional space where proximity encodes relatedness. Building the optimization machinery from scratch gave us direct insight into how these representations form and why design choices such as the loss function, corpus size, and learning rate have the effects they do, which would have been opaque had we simply called a packaged Word2Vec implementation.

miniproject

March 23, 2026

1 Dependencies

[ ]:
```
import tqdm as notebook_tqdm
from datasets import load_dataset
from dotenv import load_dotenv
from typing import Callable, Any
from pathlib import Path
import torch as T
import numpy as np
import string
import os
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
import umap
import pandas as pd
import time
```

2 Setup

[2]:
```
# Change print width
T.set_printoptions(linewidth=1000)
np.set_printoptions(linewidth=1000)
```

3 Data

Importing the data

[3]:
```
env_path = os.path.join(os.getcwd(), ".env.secrets")
load_dotenv(env_path)

hf_token = os.getenv("HF_TOKEN")

FORMAT = "torch"

ds = load_dataset(
    "Salesforce/wikitext",
```

```
    "wikitext-103-v1",
    token=hf_token if hf_token else False,
).with_format(FORMAT)

x_test, x_train, x_val = (ds[key] for key in ds.keys())

display(x_train["text"][:10])
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
```

```
['',
 ' = Valkyria Chronicles III = \n',
 '',
 ' Senjō no Valkyria 3 : <unk> Chronicles ( Japanese :      3 , lit . Valkyria␣
 ↪of the Battlefield 3 ) , commonly referred to as Valkyria Chronicles III␣
 ↪outside Japan , is a tactical role @-@ playing video game developed by Sega␣
 ↪and Media.Vision for the PlayStation Portable . Released in January 2011 in␣
 ↪Japan , it is the third game in the Valkyria series . Employing the same␣
 ↪fusion of tactical and real @-@ time gameplay as its predecessors , the story␣
 ↪runs parallel to the first game and follows the " Nameless " , a penal␣
 ↪military unit serving the nation of Gallia during the Second Europan War who␣
 ↪perform secret black operations and are pitted against the Imperial unit "␣
 ↪<unk> Raven " . \n',
 " The game began development in 2010 , carrying over a large portion of the␣
 ↪work done on Valkyria Chronicles II . While it retained the standard features␣
 ↪of the series , it also underwent multiple adjustments , such as making the␣
 ↪game more forgiving for series newcomers . Character designer <unk> Honjou and␣
 ↪composer Hitoshi Sakimoto both returned from previous entries , along with␣
 ↪Valkyria Chronicles II director Takeshi Ozawa . A large team of writers␣
 ↪handled the script . The game 's opening theme was sung by May 'n . \n",
 " It met with positive sales in Japan , and was praised by both Japanese and␣
 ↪western critics . After release , it received downloadable content , along␣
 ↪with an expanded edition in November of that year . It was also adapted into␣
 ↪manga and an original video animation series . Due to low sales of Valkyria␣
 ↪Chronicles II , Valkyria Chronicles III was not localized , but a fan␣
 ↪translation compatible with the game 's expanded edition was released in 2014 .
 ↪ Media.Vision would return to the franchise with the development of Valkyria :␣
 ↪Azure Revolution for the PlayStation 4 . \n",
 '',
 ' = = Gameplay = = \n',
 '',
```

```
" As with previous <unk> Chronicles games , Valkyria Chronicles III is a␣
↪tactical role @-@ playing game where players take control of a military unit␣
↪and take part in missions against enemy forces . Stories are told through␣
↪comic book @-@ like panels with animated character portraits , with characters␣
↪speaking partially through voiced speech bubbles and partially through␣
↪unvoiced text . The player progresses through a series of linear missions ,␣
↪gradually unlocked as maps that can be freely scanned through and replayed as␣
↪they are unlocked . The route to each story location on the map varies␣
↪depending on an individual player 's approach : when one option is selected ,␣
↪the other is sealed off to the player . Outside missions , the player␣
↪characters rest in a camp , where units can be customized and character growth␣
↪occurs . Alongside the main story missions are character @-@ specific sub␣
↪missions relating to different squad members . After the game 's completion ,␣
↪additional episodes are unlocked , some of them having a higher difficulty␣
↪than those found in the rest of the game . There are also love simulation␣
↪elements related to the game 's two main heroines , although they take a very␣
↪minor role . \n"]
```

4 Setup Training Data

[ ]:
```
class Tokenizer():
    def __init__(self, ds, num_max_sentences, window_size, PAD_TOKEN,␣
 ↪min_count=5):
        self.ds = ds
        self.num_max_sentences = num_max_sentences
        self.window_size = window_size
        self.PAD_TOKEN = PAD_TOKEN
        self.min_count = min_count

        self.vocabulary = self.create_vocabulary(ds)
        self.PAD_INDEX = len(self.vocabulary)

        self.word_to_index = {word: i for i, word in enumerate(self.vocabulary)}
        self.word_to_index[PAD_TOKEN] = self.PAD_INDEX

        self.index_to_word = {i: word for i, word in enumerate(self.vocabulary)}
        self.index_to_word[self.PAD_INDEX] = PAD_TOKEN

        self.VOCAB_SIZE = len(self.vocabulary) + 1
        self.word_freq = self._build_freq()

        print(f"Vocabulary: {self.VOCAB_SIZE} words (min_count={min_count},␣
 ↪sentences={num_max_sentences})")

    def create_vocabulary(self, data):
        """Build vocabulary from training sentences only, keeping words that
        appear at least min_count times. This keeps vocab size proportional
```

```
        to the actual training data rather than the full corpus."""
        to_remove = string.punctuation + "\u201c\u201d\u2018\u2019"
        table = str.maketrans("", "", to_remove)
        freq = {}
        for sentence in data["text"][:self.num_max_sentences]:
            for word in sentence.translate(table).lower().split():
                freq[word] = freq.get(word, 0) + 1
        return sorted(w for w, c in freq.items() if c >= self.min_count)

    def _build_freq(self) -> np.ndarray:
        freq = np.zeros(self.VOCAB_SIZE, dtype=np.float32)
        for sentence in self.ds["text"][:self.num_max_sentences]:
            for w in self.preprocess_sentence(sentence):
                idx = self.word_to_index.get(w)
                if idx is not None:
                    freq[idx] += 1
        return freq

    def get_one_hot_encoding_from_token_id(self, id: int):
        vec = T.zeros(self.VOCAB_SIZE)
        vec[id] = 1
        return vec

    def get_one_hot_encoding(self, word: str):
        vec = T.zeros(self.VOCAB_SIZE)
        vec[self.word_to_index[word]] = 1
        return vec

    def get_vector_one_hot_encodings(self, words: list[str]):
        return [self.get_one_hot_encoding(word) for word in words]

    def preprocess_sentence(self, sentence: str) -> list[str]:
        to_remove = string.punctuation + "\u201c\u201d\u2018\u2019"
        table = str.maketrans("", "", to_remove)
        return sentence.translate(table).lower().split()

    def generate_CBOW_pairs(self, add_padding=False, start=0.0, end=1.0):
        """Generator that yields (context_tensor, target_id) one pair at a time.
        Words absent from the vocabulary (below min_count) are skipped.
        start/end are fractions of num_max_sentences, used for train/val␣
↪splitting."""
        start_idx = int(start * self.num_max_sentences)
        end_idx   = int(end   * self.num_max_sentences)
        for sentence in self.ds["text"][start_idx:end_idx]:
            if len(sentence) < 1:
                continue
            # Only keep words that made it into the vocabulary
```

```
        ids = [self.word_to_index[w]
               for w in self.preprocess_sentence(sentence)
               if w in self.word_to_index]
        for center in range(len(ids)):
            context = []
            for offset in range(-self.window_size, self.window_size + 1):
                if offset == 0:
                    continue
                pos = center + offset
                if 0 <= pos < len(ids):
                    context.append(ids[pos])

            if len(context) == 0:
                continue

            if add_padding and len(context) < (2 * self.window_size):
                dn = 2 * self.window_size - len(context)
                context = [self.PAD_INDEX] * dn + context

            yield (T.tensor(context), ids[center])

def create_CBOW_pairs(
        self,
        add_padding=False,
        one_hot_encoded_context=False,
        one_hot_encoded_target=False
    ):
    pairs = []
    for sentence in (self.ds["text"][:self.num_max_sentences]):
        if len(sentence) < 1:
            continue
        ids = [self.word_to_index[w]
               for w in self.preprocess_sentence(sentence)
               if w in self.word_to_index]
        for center in range(len(ids)):
            context = []
            for offset in range(-self.window_size, self.window_size + 1):
                if offset == 0:
                    continue
                pos = center + offset
                if 0 <= pos < len(ids):
                    context.append(ids[pos])

            if len(context) == 0:
                continue

            if add_padding and len(context) < (2 * self.window_size):
```

```
                    dn = 2 * self.window_size - len(context)
                    context = [self.PAD_INDEX] * dn + context
                target = ids[center]

                if one_hot_encoded_context:
                    context = [self.get_one_hot_encoding_from_token_id(token)␣
 ↪for token in context]
                else:
                    context = T.tensor(context)

                if one_hot_encoded_target:
                    target = self.get_one_hot_encoding_from_token_id(target)

                pairs.append((context, target))
        return pairs
```

[ ]:
```
# Small corpus tokenizer - used for the computationally-expensive model␣
 ↪comparisons
# (softmax CBOW and unoptimized NS CBOW) where a small vocabulary is needed to
# keep runtimes manageable.
SMALL_TOK_FN    = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪tokenizer_small.pt"
_small_tok_path = Path(SMALL_TOK_FN)
max_sentences = 200
window_size   = 2
PAD_TOKEN     = "<PAD>"

if _small_tok_path.exists():
    tokenizer_small = T.load(_small_tok_path, weights_only=False)
    _stale_small = (
        tokenizer_small.num_max_sentences != max_sentences
        or getattr(tokenizer_small, "min_count", None) != 3
    )
    if _stale_small:
        print("Rebuilding small tokenizer …")
        tokenizer_small = Tokenizer(x_train, max_sentences, window_size,␣
 ↪PAD_TOKEN, min_count=3)
        T.save(tokenizer_small, _small_tok_path)
else:
    tokenizer_small = Tokenizer(x_train, max_sentences, window_size, PAD_TOKEN,␣
 ↪min_count=3)
    T.save(tokenizer_small, _small_tok_path)
```

[ ]:
```
class DataLoader():
    def __init__(self, data, batch_size, shuffle=False):
        self.data = data
        self.batch_size = batch_size
```

```
        self.shuffle = shuffle
        self._indices = None
        self.num_data_points = len(data)

    def __iter__(self):
        indices = list(range(len(self.data)))
        if self.shuffle:
            np.random.shuffle(indices)
        self._indices = indices
        self._pos = 0
        return self

    def __next__(self):
        if self._pos >= len(self.data):
            raise StopIteration
        batch = [self.data[i] for i in self._indices[self._pos : self._pos +␣
↪self.batch_size]]
        self._pos += self.batch_size
        return batch

    def __len__(self):
        return len(self.data) // self.batch_size


class StreamingDataLoader():
    """DataLoader that generates (context, target) pairs on demand via a␣
↪generator
    function. No list of all pairs is ever held in memory."""

    def __init__(self, pair_fn, batch_size):
        self.pair_fn = pair_fn      # callable: () -> generator of (context,␣
↪target_id)
        self.batch_size = batch_size
        # Count pairs once at construction so len() and num_data_points are␣
↪available
        self.num_data_points = sum(1 for _ in pair_fn())

    def __len__(self):
        return self.num_data_points // self.batch_size

    def __iter__(self):
        batch = []
        for pair in self.pair_fn():
            batch.append(pair)
            if len(batch) == self.batch_size:
                yield batch
                batch = []
```

```
        if batch:   # yield last partial batch
            yield batch
```

```
Train pairs: 11411  |  Val pairs: 4126
```

[ ]:
```
TRAIN_SPLIT = 0.8
# Small dataloaders - used for the showcase/comparison training runs
dl_small_train = StreamingDataLoader(
    lambda: tokenizer_small.generate_CBOW_pairs(add_padding=True,␣
 ↪end=TRAIN_SPLIT),
    batch_size=32,
)
dl_small_val = StreamingDataLoader(
    lambda: tokenizer_small.generate_CBOW_pairs(add_padding=True,␣
 ↪start=TRAIN_SPLIT),
    batch_size=32,
)
print(
    f"Small corpus  - vocab: {tokenizer_small.VOCAB_SIZE:>5}  "
    f"train pairs: {dl_small_train.num_data_points:>6}  "
    f"val pairs: {dl_small_val.num_data_points:>5}"
)
```

5 Creating the modeling tools

[ ]:
```
class Layer():
    def forward(self, v: Any) -> T.Tensor:
        raise NotImplementedError(f"{type(self).__name__} has not implemented␣
 ↪forward")
```

[ ]:
```
class Derivable(Layer):
    """A layer that can compute gradients but has no trainable weights (e.g.␣
 ↪activations)."""

    def __init__(self):
        super().__init__()
        self.dLdv: T.Tensor = None  # gradient of loss w.r.t. this layer's input

    def backward(self, dLdy: T.Tensor):
        """Compute and store dLdv given upstream gradient dLdy."""
        raise NotImplementedError(f"{type(self).__name__} has not implemented␣
 ↪backward")
```

[ ]:
```
class Trainable(Derivable):
    """A layer with trainable weights supporting mini-batch SGD.

    Mini-batch training protocol:
```

```
    1. model.zero_grad()                        - clear accumulators (once per␣
↪batch)
    2. model.accumulate_gradients(dLdy)         - per sample, inside the batch␣
↪loop
    3. model.backpropogate(batch_size)          - apply weight update (once␣
↪per batch)
   """

   def __init__(self, lr):
       super().__init__()
       self.lr:   float    = lr
       self.dLdw: T.Tensor = None  # gradient of loss w.r.t. weights
       self.dLdv: T.Tensor = None  # gradient of loss w.r.t. input

   def zero_grad(self) -> None:
       """Clear gradient accumulators before a new batch."""
       raise NotImplementedError(f"{type(self).__name__} has not implemented␣
↪zero_grad")

   def accumulate_gradients(self, dLdy: T.Tensor) -> None:
       """Compute gradients for one sample and add to accumulators.
       Sets dLdv for the chain rule. Does NOT update weights."""
       raise NotImplementedError(f"{type(self).__name__} has not implemented␣
↪accumulate_gradients")

   def update_weights(self, batch_size: int) -> None:
       """Apply accumulated gradients divided by batch_size, then reset␣
↪accumulators.
       Called once per batch after all accumulate_gradients calls."""
       raise NotImplementedError(f"{type(self).__name__} has not implemented␣
↪update_weights")
```

5.1 One-hot-encoder

Assigns each word in a vocabulary a n dimensional vector with a single element set to one (uniquely) where the words are sorted alphabetically and n is the number of unique words

[163]:
```
class OneHotEncoder(Layer):
    def __init__(self, vocabulary_size: int):
        self.dim = vocabulary_size

    def forward(self, token: int) -> T.Tensor:
        vec = T.zeros(self.dim)
        vec[token] = 1
        return vec
```

5.2 Vectorized One-hot-encoder

[164]:
```
class OneHotEncoderVectorized(Layer):
    def __init__(self, encoder: OneHotEncoder):
        self.encoder = encoder

    def forward(self, tokens: list[int]) -> T.Tensor:
        return T.stack([self.encoder.forward(token) for token in tokens])
```

[ ]:
```
FN = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/one_hot_encoder.
↪pt"
file_path = Path(FN)
if file_path.exists():
    encoder_small = T.load(file_path, weights_only=False)
    if encoder_small.dim != tokenizer_small.VOCAB_SIZE:
        print(f"Encoder stale (dim {encoder_small.dim} → {tokenizer_small.
↪VOCAB_SIZE}), rebuilding …")
        encoder_small = OneHotEncoder(tokenizer_small.VOCAB_SIZE)
        T.save(encoder_small, file_path)
else:
    encoder_small = OneHotEncoder(tokenizer_small.VOCAB_SIZE)
    T.save(encoder_small, file_path)

print(f"Encoder (small):  dim={encoder_small.dim}")
```

5.3 Softmax

[234]:
```
class Softmax(Derivable):
    def __init__(self):
        super().__init__()
        self._cached_output = None

    def forward(self, z: T.Tensor) -> T.Tensor:
        # Exponentiate all elements of z, z are the logits, or the "raw" input.
        z_exp = T.exp(z)
        # Squeeze away the last element of y, this makes no difference␣
↪mathematically.
        y = (z_exp / T.sum(z_exp, dim=0)).squeeze()

        self._cached_output = y
        return y

    def compute_gradients(self, dLdy: T.Tensor):
        return (self._cached_output * (dLdy - T.dot(self._cached_output, dLdy)))

    def backward(self, dLdy: T.Tensor):
        self.dLdv = self.compute_gradients(dLdy)
```

```
sm = Softmax()
test_values = [
    T.tensor([1.0, 0.0, 0.0, 0.0]).reshape(-1, 1),
    T.tensor([1.0, 1.0, 0.0, 0.0]).reshape(-1, 1),
    T.tensor([1.0, 1.0, 1.0, 0.0]).reshape(-1, 1),
    T.tensor([1.0, 1.0, 1.0, 1.0]).reshape(-1, 1),
]

for v in test_values:
    out = sm.forward(v)
    print(
        f"""Output:\n{out.numpy()},
        """
    )
```

```
Output:
[0.4753669 0.1748777 0.1748777 0.1748777],

Output:
[0.3655293  0.3655293  0.13447072 0.13447072],

Output:
[0.29692274 0.29692274 0.29692274 0.10923178],

Output:
[0.25 0.25 0.25 0.25],
```

5.4 Linear layer

[ ]:
```
class LinearLayer2D(Trainable):
    def __init__(self, dim_in, dim_out, activation, lr,␣
 ↪is_layer_in_average=False):
        super().__init__(lr=lr)
        self.dim_in  = dim_in
        self.dim_out = dim_out
        self.weights = T.randn(dim_in + 1, dim_out) * 0.01
        self.activation = activation
        self.is_layer_in_average = is_layer_in_average
        self._cached_input = []
        self._dLdw_accum   = T.zeros(dim_in + 1, dim_out)  # gradient␣
 ↪accumulator

    def __add_bias(self, v):
        return T.cat((v, T.ones((1,))), dim=0)
```

```
    def forward(self, v):
        vec_plus_bias = self.__add_bias(v)
        self._cached_input.append(vec_plus_bias)
        z = vec_plus_bias @ self.weights
        return self.activation.forward(z).squeeze() if self.activation else z.
↪squeeze()

    def accumulate_gradients(self, dLdy):
        """Compute gradient for this sample and accumulate. Does NOT update␣
↪weights."""
        if not self._cached_input:
            raise RuntimeError("Perform a forward pass before accumulating␣
↪gradients")
        if self.activation is not None:
            self.activation.backward(dLdy)
            dLdz = self.activation.dLdv
        else:
            dLdz = dLdy
        cached_input = self._cached_input.pop()
        self.dLdv          = (self.weights @ dLdz)[:-1]    # strip bias row
        self._dLdw_accum  += T.outer(cached_input, dLdz)

    def zero_grad(self):
        self._dLdw_accum.zero_()

    def update_weights(self, batch_size):
        """Apply accumulated gradient and reset. Call once per batch."""
        self.weights      -= self.lr * self._dLdw_accum / batch_size
        self._dLdw_accum.zero_()


test_values = [T.ones((10,), dtype=T.float32)]
ll = LinearLayer2D(10, 3, Softmax(), 1e-3)
for v in test_values:
    out = ll.forward(v)
    print(f"Input: {v.numpy()}\nOutput: {out.numpy()}\n")
```

```
Input: [1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]
Output: [0.34039378 0.33844656 0.3211597 ]
```

5.5 Linear layer average mixer

This object is used to average the the result of many layers of the same type.

[ ]:
```
class LinearLayerAvgMixer(Trainable):

    def __init__(self, n: int, layer: LinearLayer2D, lr=None):
```

```
        super().__init__(lr)
        self.n     = n
        self.layer = layer

    def forward(self, v: T.Tensor) -> T.Tensor:
        avg = T.zeros(self.layer.dim_out)
        for i in range(self.n):
            avg += self.layer.forward(v[i]) / self.n
        return avg

    def zero_grad(self) -> None:
        self.layer.zero_grad()

    def accumulate_gradients(self, dLdy: T.Tensor) -> None:
        for _ in range(self.n):
            self.layer.accumulate_gradients(dLdy / self.n)
        self.dLdv = self.layer.dLdv

    def update_weights(self, batch_size: int) -> None:
        self.layer.update_weights(batch_size)
```

5.6 Embedding Average Layer (Optimized Lookup)

The `LinearLayerAvgMixer` averages context embeddings by: 1. Encoding each context word index as a one-hot vector $e_i \in \mathbb{R}^V$ 2. Multiplying by the weight matrix: $e_i W = W_{[i]}$ (row i of W) 3. Averaging the results

Step 1 and 2 together are equivalent to a direct row lookup, since multiplying a one-hot vector by a matrix simply selects the corresponding row:

$$\frac{1}{n}\sum_{k=1}^{n} e_{i_k} W = \frac{1}{n}\sum_{k=1}^{n} W_{[i_k]}$$

`EmbeddingAverageLayer` implements this directly without large matrix multiplication and memory costly allocations. It is a mathematically equivalent simplification used to speed up training.

[ ]:
```
class EmbeddingAverageLayer(Trainable):
    """
    Optimized replacement for OneHotEncoderVectorized + LinearLayerAvgMixer.

    Mathematically equivalent to averaging one-hot context vectors multiplied␣
↪by W,
    but replaces the (vocab_size x embed_dim) matmul with a direct row lookup:

        (1/n) * sum(e_{i_k} @ W)  ==  (1/n) * sum(W[i_k])

    This avoids allocating large one-hot vectors (vocab_size-dim) entirely.
    """
```

```
    def __init__(self, vocab_size: int, embed_dim: int, lr: float = 1e-3):
        super().__init__(lr=lr)
        self.vocab_size      = vocab_size
        self.embed_dim       = embed_dim
        self.weights         = T.randn(vocab_size, embed_dim) * 0.1
        self._cached_indices = None
        self._grad_accum     = {}   # idx (int) -> accumulated gradient tensor

    def forward(self, indices: T.Tensor) -> T.Tensor:
        self._cached_indices = indices
        return self.weights[indices].mean(dim=0)   # (embed_dim,)

    def zero_grad(self) -> None:
        self._grad_accum.clear()

    def accumulate_gradients(self, dLdy: T.Tensor) -> None:
        """Accumulate gradient contribution from one sample. Does NOT update␣
↪weights."""
        n    = len(self._cached_indices)
        grad = dLdy / n
        for idx in self._cached_indices:
            idx = int(idx)
            if idx in self._grad_accum:
                self._grad_accum[idx] += grad
            else:
                self._grad_accum[idx] = grad.clone()
        self.dLdv = None   # no differentiable layer upstream

    def update_weights(self, batch_size: int) -> None:
        """Apply accumulated gradients averaged over the batch, then reset."""
        for idx, grad in self._grad_accum.items():
            self.weights[idx] -= self.lr * grad / batch_size
        self._grad_accum.clear()
```

5.7 The Model object

This object holds all the layers and activation functions we have defined, it represents the implementation of the full function that produces and outputs $\hat{\mathbf{y}}$ from an input $\mathbf{v}$.

[ ]:
```
class Model:
    def __init__(self, layers: list[Trainable | Derivable | object],␣
↪debug_mode=False):
        self.layers     = layers
        self.debug_mode = debug_mode

    def forward(self, v: T.Tensor):
```

```
        curr = v
        for l in self.layers:
            if self.debug_mode:
                print(f"Input to {type(l).__name__}: shape {curr.shape if␣
↪isinstance(curr, T.Tensor) else '?'}")
            curr = l.forward(curr)
            if self.debug_mode:
                print(f"Output from {type(l).__name__}: shape {curr.shape}")
        return curr

    def zero_grad(self) -> None:
        """Clear all gradient accumulators. Call once before each batch."""
        for l in self.layers:
            if isinstance(l, Trainable):
                l.zero_grad()

    def accumulate_gradients(self, dLdy: T.Tensor) -> None:
        """Chain the gradient backward through all layers for one sample.
        Calls accumulate_gradients on each Trainable layer"""
        for l in reversed(self.layers):
            if isinstance(l, Trainable):
                l.accumulate_gradients(dLdy)
            if not isinstance(l, OneHotEncoderVectorized):
                dLdy = l.dLdv

    def backpropogate(self, batch_size: int) -> None:
        """Apply accumulated gradients to all layers. Call once per batch."""
        for l in self.layers:
            if isinstance(l, Trainable):
                l.update_weights(batch_size)

    def clear_caches(self) -> None:
        """Discard cached forward-pass inputs after a validation forward pass.
↪"""
        for l in self.layers:
            if hasattr(l, '_cached_input'):
                l._cached_input.clear()
            if hasattr(l, 'layer') and hasattr(l.layer, '_cached_input'):
                l.layer._cached_input.clear()
```

5.8 Binary cross entropy loss

[238]:
```
class BinaryCrossEntropyLoss():
    def __init__(self):
        self.loss = None
        self.dLdy = None
```

```
    def compute_loss(self, y_true: T.Tensor, y_hat: T.Tensor):
        # Log of 0 gives inf. Clamping to avoid crashes
        y_hat = T.clamp(y_hat, min=1e-7, max=1 - 1e-7)

        loss = -(y_true @ T.log(y_hat) + (1-y_true) @ (T.log(1-y_hat)))
        self.loss = loss

        dLdy_hat = ((-y_true) / y_hat) + ((1-y_true) / (1-y_hat))
        self.dLdy = dLdy_hat
        return loss
```

5.9 Negative Sampling Loss

[ ]:
```
class NegativeSamplingLoss():
    def __init__(self, tokenizer, embed_dim, k=5, lr=1e-3, power=0.75):
        self.k              = k
        self.lr             = lr
        self.vocab_size     = tokenizer.VOCAB_SIZE
        self.out_embeddings = T.randn(self.vocab_size, embed_dim) * 0.1
        self.noise_dist     = self._build_noise_dist(tokenizer.word_freq, power)
        self._cached_h          = None
        self._cached_pos_id     = None
        self._cached_neg_ids    = None
        self._cached_pos_score  = None
        self._cached_neg_scores = None
        self.dLdv               = None
        self._pos_grad_accum    = {}   # pos_id -> accumulated gradient
        self._neg_grad_accum    = {}   # neg_id -> accumulated gradient

    def _build_noise_dist(self, word_freq, power):
        smoothed = word_freq ** power
        smoothed += 1e-10
        return smoothed / smoothed.sum()

    def _sample_negatives(self, exclude_id):
        samples = []
        while len(samples) < self.k:
            c = int(np.random.choice(self.vocab_size, p=self.noise_dist))
            if c != exclude_id:
                samples.append(c)
        return samples

    def compute_loss(self, h, pos_id):
        pos_vec   = self.out_embeddings[pos_id]
        pos_score = T.sigmoid(T.dot(h, pos_vec))
        pos_loss  = T.log(pos_score + 1e-7)
```

```
        neg_ids    = self._sample_negatives(pos_id)
        neg_scores = [T.sigmoid(T.dot(h, self.out_embeddings[i])) for i in␣
↪neg_ids]
        neg_loss   = sum(T.log(1 - s + 1e-7) for s in neg_scores)

        self._cached_h          = h
        self._cached_pos_id     = pos_id
        self._cached_neg_ids    = neg_ids
        self._cached_pos_score  = pos_score
        self._cached_neg_scores = neg_scores
        return -(pos_loss + neg_loss)

    def zero_grad(self) -> None:
        self._pos_grad_accum.clear()
        self._neg_grad_accum.clear()

    def backward(self) -> None:
        """Compute dLdv for chain rule and accumulate output embedding␣
↪gradients.
        Does NOT update weights"""
        h         = self._cached_h
        pos_id    = self._cached_pos_id
        pos_score = self._cached_pos_score

        # Gradient w.r.t. h - passed back to input embedding layer via dLdv
        dLdh = (pos_score - 1) * self.out_embeddings[pos_id].clone()

        pos_grad = (pos_score - 1) * h
        if pos_id in self._pos_grad_accum:
            self._pos_grad_accum[pos_id] += pos_grad
        else:
            self._pos_grad_accum[pos_id] = pos_grad.clone()

        for neg_id, neg_score in zip(self._cached_neg_ids, self.
↪_cached_neg_scores):
            dLdh += neg_score * self.out_embeddings[neg_id].clone()
            neg_grad = neg_score * h
            if neg_id in self._neg_grad_accum:
                self._neg_grad_accum[neg_id] += neg_grad
            else:
                self._neg_grad_accum[neg_id] = neg_grad.clone()

        self.dLdv = dLdh

    def update_weights(self, batch_size: int) -> None:
        """Apply accumulated gradients averaged over the batch, then reset."""
```

```
        for idx, grad in self._pos_grad_accum.items():
            self.out_embeddings[idx] -= self.lr * grad / batch_size
        for idx, grad in self._neg_grad_accum.items():
            self.out_embeddings[idx] -= self.lr * grad / batch_size
        self._pos_grad_accum.clear()
        self._neg_grad_accum.clear()
```

6 Creating the Word2Vec model

[ ]:
```
# Softmax CBOW - trained on the small corpus
# Full softmax requires a dense (LATENT_DIM × VOCAB_SIZE) output projection and
# computes a probability distribution over the ENTIRE vocabulary on every␣
 ↪forward
# and backward pass: O(V) cost per sample.  This becomes the dominant bottleneck
# as V grows

INPUT_DIM  = encoder_small.dim   # = tokenizer_small.VOCAB_SIZE
OUTPUT_DIM = encoder_small.dim
LATENT_DIM = 10

word2vecCBOW = Model(
    [
        OneHotEncoderVectorized(encoder=encoder_small),
        LinearLayerAvgMixer(
            n=4,
            layer=LinearLayer2D(
                dim_in=INPUT_DIM,
                dim_out=LATENT_DIM,
                activation=None,
                lr=1e-3,
                is_layer_in_average=True,
            ),
        ),
        LinearLayer2D(
            dim_in=LATENT_DIM,
            dim_out=OUTPUT_DIM,
            activation=Softmax(),
            lr=1e-3,
            is_layer_in_average=False,
        ),
    ],
    debug_mode=False,
)

_n_params_softmax = INPUT_DIM * LATENT_DIM + LATENT_DIM * OUTPUT_DIM
print(f"Softmax CBOW | vocab: {INPUT_DIM} | latent: {LATENT_DIM}")
```

```
print(f"  Trainable params : {_n_params_softmax:,}")
print(f"  Output cost/step : O(V={OUTPUT_DIM}) - full softmax normalisation")
```

6.1 Negative Sampling Word2Vec

[ ]:
```
# Unoptimized NS CBOW - trained on the small corpus
# Negative sampling replaces the O(V) softmax output with O(k) comparisons, but
# the INPUT side still uses full one-hot context vectors and a dense
# (V × LATENT_DIM) matrix multiply.  That input projection is also O(V) per
# context word, so the speedup over softmax is limited to the output layer only.
# The EmbeddingAverageLayer (fast model) eliminates this by replacing the dense
# matmul with a direct O(1) row lookup.

INPUT_DIM   = encoder_small.dim   # = tokenizer_small.VOCAB_SIZE
LATENT_DIM  = 10
K_NEGATIVES = 5

word2vecCBOW_ns = Model(
    [
        OneHotEncoderVectorized(encoder=encoder_small),
        LinearLayerAvgMixer(
            n=4,
            layer=LinearLayer2D(
                dim_in=INPUT_DIM,
                dim_out=LATENT_DIM,
                activation=None,
                lr=1e-3,
                is_layer_in_average=True,
            ),
        ),
    ],
    debug_mode=False,
)

ns_loss = NegativeSamplingLoss(
    tokenizer=tokenizer_small,
    embed_dim=LATENT_DIM,
    k=K_NEGATIVES,
    lr=1e-3,
)

_n_params_ns = INPUT_DIM * LATENT_DIM + tokenizer_small.VOCAB_SIZE * LATENT_DIM
print(f"NS CBOW (unoptimized) | vocab: {INPUT_DIM} | latent: {LATENT_DIM} |␣
↪k={K_NEGATIVES}")
print(f"  Trainable params  : {_n_params_ns:,}")
print(f"  Input cost/step   : O(V={INPUT_DIM}) - dense one-hot × weight matrix")
print(f"  Output cost/step  : O(k={K_NEGATIVES}) - negative sampling")
```

7 Training, Validation, Testing

7.1 Training harness

Two different training loops are required for binary cross entropy loss and negative sampling loss respectively since the negative sampling loss has trainable weights while not fitting explicitly into the model definition. Therefore, the update is done inside the training loop instead if as part of the model backpropogation method. This could be improved as part of future work.

[ ]:
```
def train(model, dataloader, val_dataloader, loss_fn, tokenizer, n_epochs):
    train_losses = []
    val_losses   = []

    for epoch in range(n_epochs):
        # --- Training ---
        epoch_train_loss = 0
        pbar = notebook_tqdm.tqdm(dataloader, total=len(dataloader),␣
↪desc=f"Epoch {epoch+1}/{n_epochs}", leave=True)
        for batch in pbar:
            model.zero_grad()
            batch_loss = 0

            for (context, target) in batch:
                target_one_hot = tokenizer.
↪get_one_hot_encoding_from_token_id(int(target))
                y_hat = model.forward(context)
                loss  = loss_fn.compute_loss(target_one_hot, y_hat)
                batch_loss       += loss
                model.accumulate_gradients(loss_fn.dLdy)    # accumulate per␣
↪sample

            model.backpropogate(len(batch))                 # weight update once␣
↪per batch
            epoch_train_loss += batch_loss
            pbar.set_postfix({"train_loss": f"{float(epoch_train_loss):.4f}"})

        epoch_train_loss /= dataloader.num_data_points
        train_losses.append(epoch_train_loss)

        # --- Validation (no weight updates) ---
        epoch_val_loss = 0
        for batch in val_dataloader:
            for (context, target) in batch:
                target_one_hot = tokenizer.
↪get_one_hot_encoding_from_token_id(int(target))
                y_hat = model.forward(context)
                model.clear_caches()
                epoch_val_loss += loss_fn.compute_loss(target_one_hot, y_hat)
```

```
        epoch_val_loss /= val_dataloader.num_data_points
        val_losses.append(epoch_val_loss)

        print(f"Epoch {epoch+1}/{n_epochs}: train={float(epoch_train_loss):.4f}␣
↪ val={float(epoch_val_loss):.4f}")

    return {"train": train_losses, "val": val_losses}
```

[ ]:
```
def train_ns(model, ns_loss, dataloader, val_dataloader, n_epochs):
    train_losses = []
    val_losses   = []

    for epoch in range(n_epochs):
        # --- Training ---
        epoch_train_loss = 0
        pbar = notebook_tqdm.tqdm(dataloader, total=len(dataloader),␣
↪desc=f"Epoch {epoch+1}/{n_epochs}", leave=True)
        for batch in pbar:
            model.zero_grad()
            ns_loss.zero_grad()
            batch_loss = 0

            for (context, target_id) in batch:
                target_id = int(target_id)
                h    = model.forward(context)
                loss = ns_loss.compute_loss(h, target_id)
                batch_loss += loss.item()
                ns_loss.backward()                          # accumulate per␣
↪sample
                model.accumulate_gradients(ns_loss.dLdv)   # accumulate per␣
↪sample

            ns_loss.update_weights(len(batch))              # weight update␣
↪once per batch
            model.backpropogate(len(batch))                 # weight update␣
↪once per batch

            epoch_train_loss += batch_loss
            pbar.set_postfix({"train_loss": f"{epoch_train_loss:.4f}"})

        epoch_train_loss /= dataloader.num_data_points
        train_losses.append(epoch_train_loss)

        # --- Validation (no weight updates) ---
        epoch_val_loss = 0
        for batch in val_dataloader:
```

```
            for (context, target_id) in batch:
                target_id = int(target_id)
                h     = model.forward(context)
                model.clear_caches()
                epoch_val_loss += ns_loss.compute_loss(h, target_id).item()

        epoch_val_loss /= val_dataloader.num_data_points
        val_losses.append(epoch_val_loss)

        print(f"Epoch {epoch+1}/{n_epochs}: train={epoch_train_loss:.4f} ␣
↪val={epoch_val_loss:.4f}")

    return {"train": train_losses, "val": val_losses}
```

7.2 Continous bag of words

[ ]:
```
ce = BinaryCrossEntropyLoss()

_t0 = time.perf_counter()
softmax_history = train(
    word2vecCBOW, dl_small_train, dl_small_val, ce, tokenizer_small, n_epochs=5
)
_softmax_time = time.perf_counter() - _t0
print(f"\nSoftmax CBOW - 5 epochs | {_softmax_time:.1f}s total  "
      f"({_softmax_time / dl_small_train.num_data_points / 5 * 1000:.2f} ms/
↪sample)")
```

```
Epoch 1/5: 176it [00:02, 87.35it/s, train_loss=41114.2188]

Epoch 1/5: train=7.3157  val=7.3182

Epoch 2/5: 176it [00:02, 83.40it/s, train_loss=41095.5859]

Epoch 2/5: train=7.3124  val=7.3151

Epoch 3/5: 176it [00:02, 85.59it/s, train_loss=41076.9609]

Epoch 3/5: train=7.3091  val=7.3121

Epoch 4/5: 176it [00:02, 84.63it/s, train_loss=41058.3672]

Epoch 4/5: train=7.3058  val=7.3090

Epoch 5/5: 176it [00:02, 82.90it/s, train_loss=41039.7734]

Epoch 5/5: train=7.3025  val=7.3060

Softmax CBOW - 5 epochs | 11.2s total  (0.40 ms/sample)
```

[ ]:
```
_t0 = time.perf_counter()
ns_history = train_ns(
    word2vecCBOW_ns, ns_loss, dl_small_train, dl_small_val, n_epochs=5
```

```
)
_ns_time = time.perf_counter() - _t0
print(f"\nNS CBOW (unoptimized) - 5 epochs | {_ns_time:.1f}s total  "
      f"({_ns_time / dl_small_train.num_data_points / 5 * 1000:.2f} ms/sample)")
```

7.3 Optimized Word2Vec with EmbeddingAverageLayer

The model below uses `EmbeddingAverageLayer` instead of `OneHotEncoderVectorized + LinearLayerAvgMixer`. The architecture is identical in terms of what it learns where the input is now the raw integer context tensor rather than one-hot vectors.

[ ]:
```
LATENT_DIM_FAST  = 50
K_NEGATIVES_FAST = 5
LR               = 0.2
N_EPOCHS         = 10

embedding_layer = EmbeddingAverageLayer(
    vocab_size=tokenizer_small.VOCAB_SIZE,
    embed_dim=LATENT_DIM_FAST,
    lr=LR
)
word2vecCBOW_fast = Model([embedding_layer], debug_mode=False)

ns_loss = NegativeSamplingLoss(
    tokenizer=tokenizer_small,
    embed_dim=LATENT_DIM_FAST,
    k=K_NEGATIVES_FAST,
    lr=LR
)

_t0 = time.perf_counter()
ns_fast_history = train_ns(
    word2vecCBOW_fast, ns_loss, dl_small_train, dl_small_val, n_epochs=5
)
_ns_optimized_time = time.perf_counter() - _t0
print(f"\nNS CBOW (optimized) - 5 epochs | {_ns_optimized_time:.1f}s total  "
      f"({_ns_optimized_time / dl_small_train.num_data_points / 5 * 1000:.2f}␣
 ↪ms/sample)")
```

```
Epoch 1/5: 176it [00:02, 69.16it/s, train_loss=23351.6172]

Epoch 1/5: train=4.1551  val=4.1504

Epoch 2/5: 176it [00:02, 70.42it/s, train_loss=23233.6503]

Epoch 2/5: train=4.1341  val=4.1202

Epoch 3/5: 176it [00:02, 72.88it/s, train_loss=22962.8649]

Epoch 3/5: train=4.0859  val=4.0572
```

```
Epoch 4/5: 176it [00:02, 69.88it/s, train_loss=22409.9073]

Epoch 4/5: train=3.9875  val=3.9675

Epoch 5/5: 176it [00:02, 70.91it/s, train_loss=21680.5903]

Epoch 5/5: train=3.8578  val=3.8233

NS CBOW (optimized) - 5 epochs | 13.7s total  (0.49 ms/sample)
```

[ ]:
```
# Complexity comparison
V  = tokenizer_small.VOCAB_SIZE
d  = LATENT_DIM
k  = K_NEGATIVES
ms_softmax = _softmax_time / (dl_small_train.num_data_points * 5) * 1000
ms_ns = _ns_time / (dl_small_train.num_data_points * 5) * 1000
ms_ns_optimized = _ns_optimized_time / (dl_small_train.num_data_points * 5) *␣
↪1000

print("=" * 68)
print("  Per-Sample Cost Breakdown (small corpus, 5 epochs)")
print("=" * 68)
print(f"  {'Model':<36}  {'ms/sample':>10}  {'Relative speed':>14}")
print(f"  {' '*36}  {' '*10}  {' '*14}")
print(f"  {'Softmax CBOW':<36}  {ms_softmax:>9.3f}  {'1×':>14}")
print(f"  {'NS CBOW (unoptimized)':<36}  {ms_ns:>9.3f}  {ms_softmax/ms_ns:>13.
↪2f}×")
print(f"  {'NS CBOW (optimized)':<36}  {ms_ns_optimized:>9.3f}  {ms_softmax/
↪ms_ns_optimized:>13.2f}×")
```

```
====================================================================
  Per-Sample Cost Breakdown (small corpus, 5 epochs)
====================================================================
  Model                                  ms/sample  Relative speed

  Softmax CBOW                               0.420              1×
  NS CBOW (unoptimized)                      0.648           0.65×
  NS CBOW (optimized)                        0.486           0.86×
```

7.4 Multi-Scale Training

The same `EmbeddingAverageLayer` + negative-sampling architecture is trained on three corpus sizes to study how data volume affects embedding quality. All variants share the same hyper-parameters (`embed_dim=50`, `k=5` negatives, `lr=0.2`, `batch_size=32`, 10 epochs) so that only the training data volume differs.

Variant	Sentences	min_count	Vocab Size
Small	200	3	~600
Medium	1 000	3	~2600

Variant	Sentences	min_count	Vocab Size
Large	5 000	5	~5 500

Checkpoints are saved after the first run and reloaded on subsequent runs, so re-executing the notebook does not retrain.

[ ]:
```
WEIGHTS_SMALL = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪word2vec_ns_small.pt"
_small_weights_path = Path(WEIGHTS_SMALL)

embedding_layer_small = EmbeddingAverageLayer(
    vocab_size=tokenizer_small.VOCAB_SIZE,
    embed_dim=LATENT_DIM_FAST,
    lr=LR,
)
word2vec_small = Model([embedding_layer_small], debug_mode=False)
ns_loss_small  = NegativeSamplingLoss(
    tokenizer=tokenizer_small,
    embed_dim=LATENT_DIM_FAST,
    k=K_NEGATIVES_FAST,
    lr=LR,
)

if _small_weights_path.exists():
    ckpt = T.load(_small_weights_path, weights_only=False)
    embedding_layer_small.weights = ckpt["input_embeddings"]
    ns_loss_small.out_embeddings  = ckpt["output_embeddings"]
    ns_history_small              = ckpt["history"]
    print(f"Loaded small model  ({len(ns_history_small['train'])} epochs␣
 ↪trained)")
else:
    ns_history_small = train_ns(
        word2vec_small, ns_loss_small, dl_small_train, dl_small_val, N_EPOCHS
    )
    T.save({
        "input_embeddings":  embedding_layer_small.weights,
        "output_embeddings": ns_loss_small.out_embeddings,
        "history":           ns_history_small,
    }, _small_weights_path)
    print(f"Saved small model to {WEIGHTS_SMALL}")
```

```
Loaded small model  (10 epochs trained)
```

[ ]:
```
MEDIUM_TOK_FN    = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪tokenizer_medium.pt"
_medium_tok_path = Path(MEDIUM_TOK_FN)
medium_max_num_sentences = 1000
```

```
if _medium_tok_path.exists():
    tokenizer_medium = T.load(_medium_tok_path, weights_only=False)
    _stale_medium = (
        tokenizer_medium.num_max_sentences != medium_max_num_sentences
        or getattr(tokenizer_medium, "min_count", None) != 3
    )
    if _stale_medium:
        print("Rebuilding medium tokenizer …")
        tokenizer_medium = Tokenizer(x_train, medium_max_num_sentences,␣
 ↪window_size, PAD_TOKEN, min_count=3)
        T.save(tokenizer_medium, _medium_tok_path)
else:
    tokenizer_medium = Tokenizer(x_train, medium_max_num_sentences,␣
 ↪window_size, PAD_TOKEN, min_count=3)
    T.save(tokenizer_medium, _medium_tok_path)

dl_medium_train = StreamingDataLoader(
    lambda: tokenizer_medium.generate_CBOW_pairs(add_padding=True,␣
 ↪end=TRAIN_SPLIT),
    batch_size=32,
)
dl_medium_val = StreamingDataLoader(
    lambda: tokenizer_medium.generate_CBOW_pairs(add_padding=True,␣
 ↪start=TRAIN_SPLIT),
    batch_size=32,
)
print(
    f"Medium corpus | vocab: {tokenizer_medium.VOCAB_SIZE:>2} | "
    f"train pairs: {dl_medium_train.num_data_points:>2} | "
    f"val pairs: {dl_medium_val.num_data_points:>2}"
)
```

```
Rebuilding medium tokenizer …
Vocabulary: 2588 words (min_count=3, sentences=1000)
Medium corpus | vocab: 2588 | train pairs: 31190 | val pairs: 8714
```

[ ]:
```
WEIGHTS_MEDIUM = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪word2vec_ns_medium.pt"
_medium_weights_path = Path(WEIGHTS_MEDIUM)

embedding_layer_medium = EmbeddingAverageLayer(
    vocab_size=tokenizer_medium.VOCAB_SIZE,
    embed_dim=LATENT_DIM_FAST,
    lr=LR,
)
word2vec_medium = Model([embedding_layer_medium], debug_mode=False)
```

```
ns_loss_medium  = NegativeSamplingLoss(
    tokenizer=tokenizer_medium,
    embed_dim=LATENT_DIM_FAST,
    k=K_NEGATIVES_FAST,
    lr=LR,
)

if _medium_weights_path.exists():
    ckpt = T.load(_medium_weights_path, weights_only=False)
    embedding_layer_medium.weights = ckpt["input_embeddings"]
    ns_loss_medium.out_embeddings  = ckpt["output_embeddings"]
    ns_history_medium              = ckpt["history"]
    print(f"Loaded medium model  ({len(ns_history_medium['train'])} epochs␣
 ↪trained)")
else:
    ns_history_medium = train_ns(
        word2vec_medium, ns_loss_medium, dl_medium_train, dl_medium_val,␣
 ↪N_EPOCHS
    )
    T.save({
        "input_embeddings":  embedding_layer_medium.weights,
        "output_embeddings": ns_loss_medium.out_embeddings,
        "history":           ns_history_medium,
    }, _medium_weights_path)
    print(f"Saved medium model to {WEIGHTS_MEDIUM}")
```

```
Loaded medium model  (10 epochs trained)
```

[ ]:
```
LARGE_TOK_FN    = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪tokenizer_large.pt"
_large_tok_path = Path(LARGE_TOK_FN)
large_max_num_sentences = 5000

if _large_tok_path.exists():
    tokenizer_large = T.load(_large_tok_path, weights_only=False)
    _stale_large = (
        tokenizer_large.num_max_sentences != large_max_num_sentences
        or getattr(tokenizer_large, "min_count", None) != 5
    )
    if _stale_large:
        print("Rebuilding large tokenizer …")
        tokenizer_large = Tokenizer(x_train, large_max_num_sentences,␣
 ↪window_size, PAD_TOKEN, min_count=5)
        T.save(tokenizer_large, _large_tok_path)
else:
    tokenizer_large = Tokenizer(x_train, large_max_num_sentences, window_size,␣
 ↪PAD_TOKEN, min_count=5)
```

```
    T.save(tokenizer_large, _large_tok_path)

dl_large_train = StreamingDataLoader(
    lambda: tokenizer_large.generate_CBOW_pairs(add_padding=True,␣
 ↪end=TRAIN_SPLIT),
    batch_size=32,
)
dl_large_val = StreamingDataLoader(
    lambda: tokenizer_large.generate_CBOW_pairs(add_padding=True,␣
 ↪start=TRAIN_SPLIT),
    batch_size=32,
)
print(
    f"Large corpus | vocab: {tokenizer_large.VOCAB_SIZE:>2} | "
    f"train pairs: {dl_large_train.num_data_points:>2} | "
    f"val pairs: {dl_large_val.num_data_points:>2}"
)
```

```
Rebuilding large tokenizer …
Vocabulary: 5576 words (min_count=5, sentences=5000)
Large corpus | vocab: 5576 | train pairs: 172850 | val pairs: 33148
```

[ ]:
```
WEIGHTS_LARGE = "/home/enarde/Documents/Design-Of-AI-Systems/miniproject/
 ↪word2vec_ns_large.pt"
_large_weights_path = Path(WEIGHTS_LARGE)

embedding_layer_large = EmbeddingAverageLayer(
    vocab_size=tokenizer_large.VOCAB_SIZE,
    embed_dim=LATENT_DIM_FAST,
    lr=LR,
)
word2vec_large = Model([embedding_layer_large], debug_mode=False)
ns_loss_large  = NegativeSamplingLoss(
    tokenizer=tokenizer_large,
    embed_dim=LATENT_DIM_FAST,
    k=K_NEGATIVES_FAST,
    lr=LR,
)

if _large_weights_path.exists():
    ckpt = T.load(_large_weights_path, weights_only=False)
    embedding_layer_large.weights = ckpt["input_embeddings"]
    ns_loss_large.out_embeddings  = ckpt["output_embeddings"]
    ns_history_large              = ckpt["history"]
    print(f"Loaded large model  ({len(ns_history_large['train'])} epochs␣
 ↪trained)")
else:
```

```python
ns_history_large = train_ns(
    word2vec_large, ns_loss_large, dl_large_train, dl_large_val, N_EPOCHS
)
T.save({
    "input_embeddings":  embedding_layer_large.weights,
    "output_embeddings": ns_loss_large.out_embeddings,
    "history":           ns_history_large,
}, _large_weights_path)
print(f"Saved large model to {WEIGHTS_LARGE}")
```

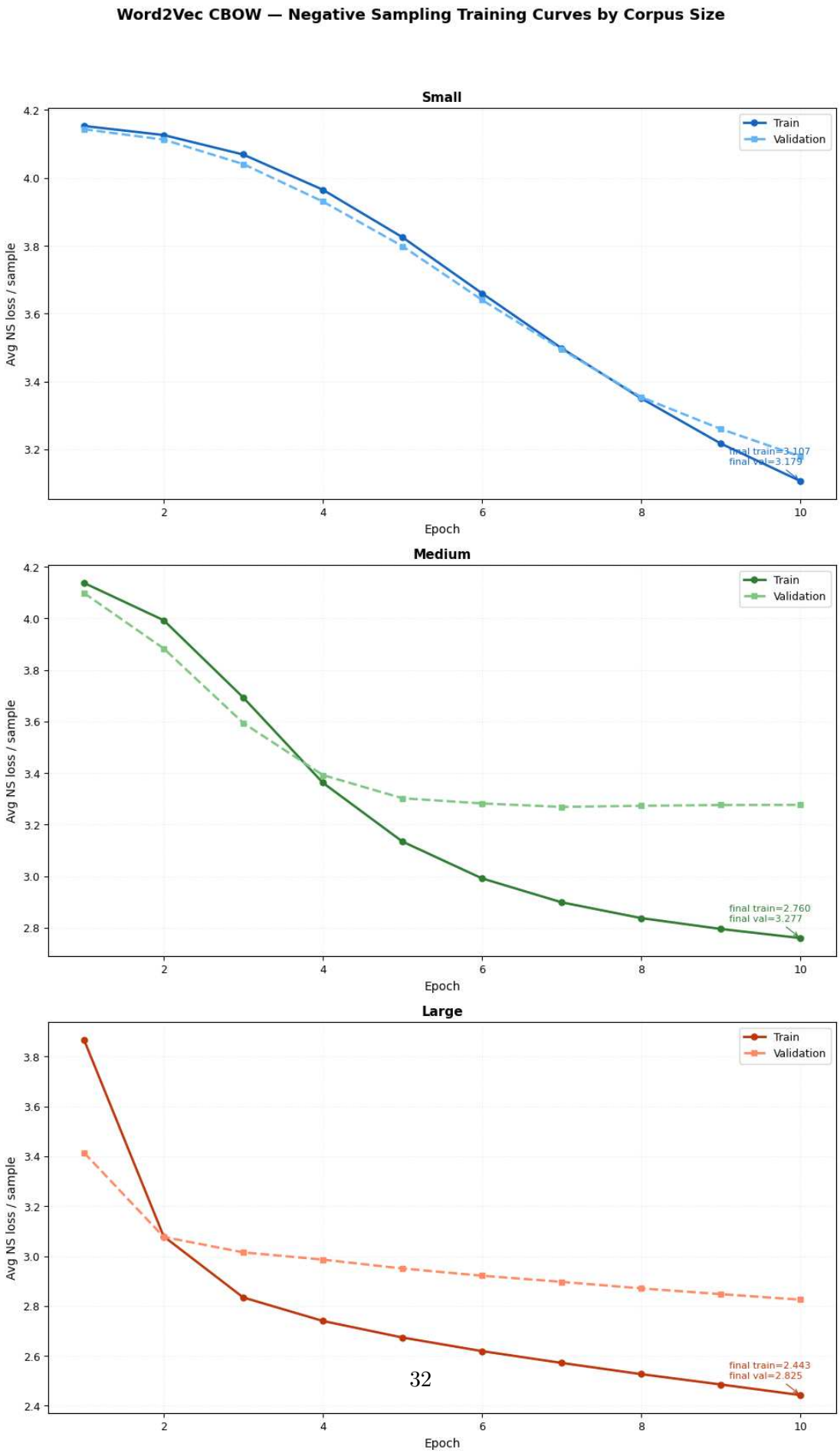
```
Epoch 1/10: 5402it [01:39, 54.06it/s, train_loss=668270.0683]

Epoch 1/10: train=3.8662  val=3.4144

Epoch 2/10: 5402it [01:41, 52.96it/s, train_loss=532166.0391]

Epoch 2/10: train=3.0788  val=3.0767

Epoch 3/10: 5402it [01:42, 52.58it/s, train_loss=489858.1849]

Epoch 3/10: train=2.8340  val=3.0148

Epoch 4/10: 5402it [01:40, 53.91it/s, train_loss=473578.2279]

Epoch 4/10: train=2.7398  val=2.9858

Epoch 5/10: 5402it [01:42, 52.86it/s, train_loss=462101.3629]

Epoch 5/10: train=2.6734  val=2.9502

Epoch 6/10: 5402it [01:44, 51.57it/s, train_loss=452676.5024]

Epoch 6/10: train=2.6189  val=2.9212

Epoch 7/10: 5402it [01:44, 51.75it/s, train_loss=444506.1285]

Epoch 7/10: train=2.5716  val=2.8966

Epoch 8/10: 5402it [01:43, 52.18it/s, train_loss=436723.9294]

Epoch 8/10: train=2.5266  val=2.8704

Epoch 9/10: 5402it [01:43, 52.19it/s, train_loss=429566.5759]

Epoch 9/10: train=2.4852  val=2.8474

Epoch 10/10: 5402it [01:40, 53.70it/s, train_loss=422200.3557]

Epoch 10/10: train=2.4426  val=2.8251
Saved large model to /home/enarde/Documents/Design-Of-AI-
Systems/miniproject/word2vec_ns_large.pt
```

7.5 Visualizing results

7.5.1 Training vs Validation Loss Plot

[ ]:
```
# Combined training-curve comparison
_scale_configs = [
    ("Small",  "ns_history_small"),
    ("Medium", "ns_history_medium"),
    ("Large", "ns_history_large"),
]
_palette = [("#1565C0", "#64B5F6"), ("#2E7D32", "#81C784"), ("#BF360C",␣
 ↪"#FF8A65")]

# Only include variants that have been trained
_histories = []
for label, var_name in _scale_configs:
    try:
        _histories.append((label, eval(var_name)))
    except NameError:
        pass

n_plots = len(_histories)
fig, axes = plt.subplots(n_plots, 1, figsize=(10, 5.5 * n_plots), sharey=False)
if n_plots == 1:
    axes = [axes]

for ax, (label, hist), (c_tr, c_val) in zip(axes, _histories, _palette):
    epochs = range(1, len(hist["train"]) + 1)
    ax.plot(epochs, hist["train"], color=c_tr,  marker="o", lw=2,
            label="Train",      ms=5)
    ax.plot(epochs, hist["val"],   color=c_val, marker="s", lw=2,
            label="Validation", ms=5, linestyle="--")
    ax.set_title(label, fontsize=11, fontweight="bold")
    ax.set_xlabel("Epoch", fontsize=10)
    ax.set_ylabel("Avg NS loss / sample", fontsize=10)
    ax.legend(fontsize=9)
    ax.grid(True, alpha=0.25, linestyle=":")
    ax.tick_params(labelsize=9)
    final_tr  = hist["train"][-1]
    final_val = hist["val"][-1]
    ax.annotate(
        f"final train={final_tr:.3f}\nfinal val={final_val:.3f}",
        xy=(len(epochs), final_tr),
        xytext=(-60, 14),
        textcoords="offset points",
        fontsize=8,
        color=c_tr,
        arrowprops=dict(arrowstyle="->", color=c_tr, lw=0.8),
```

```
    )

fig.suptitle(
    "Word2Vec CBOW - Negative Sampling Training Curves by Corpus Size",
    fontsize=13, fontweight="bold", y=1.03,
)
plt.tight_layout()
plt.show()
```


Word2Vec CBOW — Negative Sampling Training Curves by Corpus Size
Small
Train
Validation
final train=3.107
final val=3.179
Avg NS loss / sample
Epoch
Medium
Train
Validation
final train=2.760
final val=3.277
Avg NS loss / sample
Epoch
Large
Train
Validation
final train=2.443
final val=2.825
Avg NS loss / sample
Epoch

7.5.2 K-Nearest Neighbors (Cosine Similarity)

[ ]:
```
def nearest_neighbors(word, embeddings, word_to_index, index_to_word, n=8):
    """Return the n most similar words by cosine similarity."""
    idx = word_to_index.get(word)
    if idx is None:
        return None
    vec    = embeddings[idx]
    norms  = embeddings.norm(dim=1).clamp(min=1e-8)
    scores = (embeddings @ vec) / (norms * vec.norm().clamp(min=1e-8))
    top    = scores.argsort(descending=True)[1 : n + 1]   # skip self at rank 0
    return [(index_to_word[i.item()], round(scores[i].item(), 4)) for i in top]


def neighbors_dataframe(probe_words, embeddings, word_to_index, index_to_word,␣
↪n=5):
    """Build a DataFrame with one row per probe word and columns for each rank.
↪"""
    rows = []
    for word in probe_words:
        result = nearest_neighbors(word, embeddings, word_to_index,␣
↪index_to_word, n=n)
        if result is None:
            continue
        row = {"Query": word}
        for rank, (nbr, score) in enumerate(result, 1):
            row[f"#{rank}"] = f"{nbr}  ({score:.3f})"
        rows.append(row)
    return pd.DataFrame(rows).set_index("Query")


# Probe words grouped by semantic category
probe_words = [
    # time / sequence
    "time", "year", "century", "age",
    # geography / politics
    "world", "city", "war", "government", "american",
    # people / society
    "people", "king", "queen", "church",
    # descriptive
    "new", "great", "first", "history",
]

print("=" * 72)
```

```python
print("  Nearest Neighbours - Large corpus")
print("=" * 72)
df_medium = neighbors_dataframe(
    probe_words,
    embedding_layer_large.weights,
    tokenizer_large.word_to_index,
    tokenizer_large.index_to_word,
    n=8,
)
display(df_medium)

# Cross-scale comparison for a subset of words
_compare_words = ["time", "world", "war", "king", "first"]
_models = [
    ("Small",  embedding_layer_small, tokenizer_small),
    ("Medium",  embedding_layer_medium, tokenizer_medium),
    ("Large", embedding_layer_large, tokenizer_large),
]

print("\n" + "=" * 72)
print("  Cross-Scale Comparison - top-5 neighbours per model")
print("=" * 72)

for query in _compare_words:
    print(f"\n  Query: '{query}'")
    print(f"  {'Model':<22}  Neighbours")
    print(f"  {' '*22}  {' '*45}")
    for model_label, emb_layer, tok in _models:
        result = nearest_neighbors(
            query, emb_layer.weights, tok.word_to_index, tok.index_to_word, n=5
        )
        if result is None:
            print(f"  {model_label:<22}  (word not in vocabulary)")
        else:
            nbrs = ",  ".join(f"{w} ({s:.3f})" for w, s in result)
            print(f"  {model_label:<22}  {nbrs}")
```

```
========================================================================
  Nearest Neighbours - Large corpus
========================================================================

                              #1                  #2                #3  \
Query
time            british  (0.906)       day  (0.893)     same  (0.887)
year               only  (0.885)       but  (0.877)     time  (0.856)
century       following  (0.860)    before  (0.854)    under  (0.845)
age         archaeology  (0.741)      best  (0.728)       ga  (0.727)
world              game  (0.872)   british  (0.853)   second  (0.850)
```

city	first	(0.888)	top	(0.877)	team	(0.864)
war	season	(0.870)	new	(0.866)	british	(0.863)
government	team	(0.856)	gods	(0.839)	song	(0.836)
american	game	(0.896)	season	(0.878)	league	(0.875)
people	an	(0.876)	out	(0.871)	many	(0.865)
king	project	(0.827)	season	(0.753)	team	(0.753)
queen	player	(0.641)	aviation	(0.576)	rays	(0.562)
church	film	(0.856)	album	(0.826)	ship	(0.819)
new	second	(0.912)	first	(0.908)	season	(0.893)
great	side	(0.784)	third	(0.777)	made	(0.773)
first	second	(0.936)	season	(0.917)	new	(0.908)
history	some	(0.879)	first	(0.877)	games	(0.868)

	#4		#5		#6	\
Query						
time	team	(0.883)	first	(0.881)	end	(0.870)
year	new	(0.853)	egyptian	(0.853)	british	(0.850)
century	at	(0.840)	during	(0.829)	against	(0.828)
age	number	(0.722)	four	(0.721)	university	(0.721)
world	early	(0.833)	league	(0.830)	american	(0.827)
city	end	(0.857)	second	(0.851)	british	(0.851)
war	first	(0.860)	all	(0.849)	end	(0.849)
government	army	(0.834)	site	(0.830)	film	(0.826)
american	only	(0.867)	british	(0.867)	national	(0.865)
people	some	(0.859)	their	(0.856)	these	(0.854)
king	rock	(0.752)	building	(0.750)	area	(0.750)
queen	as	(0.560)	for	(0.553)	full	(0.549)
church	second	(0.817)	project	(0.811)	team	(0.807)
new	british	(0.891)	only	(0.890)	best	(0.887)
great	main	(0.767)	club	(0.763)	season	(0.761)
first	team	(0.904)	best	(0.903)	film	(0.896)
history	north	(0.865)	its	(0.863)	all	(0.861)

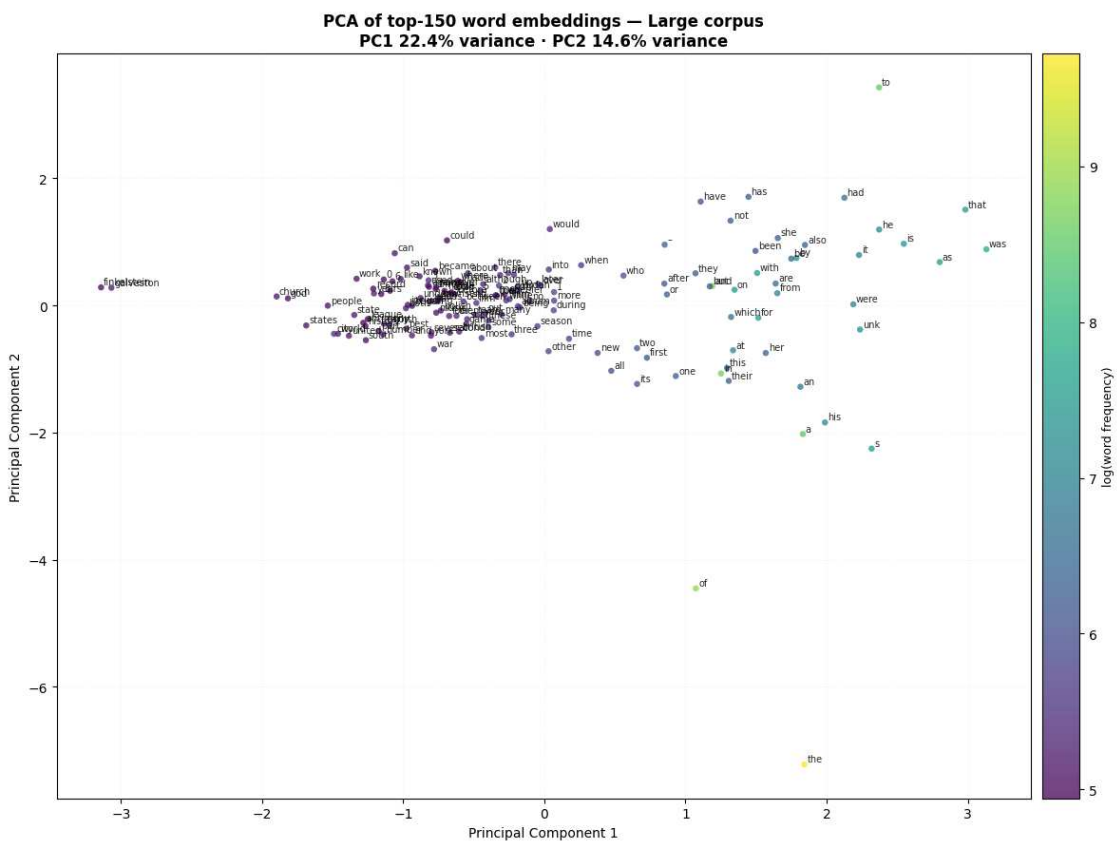
	#7		#8	
Query				
time	north	(0.870)	new	(0.867)
year	made	(0.848)	half	(0.847)
century	from	(0.825)	team	(0.823)
age	out	(0.717)	townsend	(0.716)
world	top	(0.825)	school	(0.822)
city	site	(0.851)	season	(0.848)
war	team	(0.849)	state	(0.846)
government	fleet	(0.826)	album	(0.825)
american	three	(0.863)	north	(0.860)
people	each	(0.851)	all	(0.851)
king	although	(0.747)	before	(0.744)
queen	association	(0.546)	or	(0.541)
church	song	(0.807)	last	(0.806)

```
new                 two  (0.885)       all  (0.883)
great             album  (0.758)    school  (0.755)
first              york  (0.895)       all  (0.894)
history           their  (0.857)       new  (0.852)


======================================================================
  Cross-Scale Comparison - top-5 neighbours per model
======================================================================

  Query: 'time'
  Model                   Neighbours

  Small                   mary (0.398),  available (0.391),  over (0.371),
major (0.354),  removed (0.353)
  Medium                  in (0.817),  at (0.791),  and (0.782),  was (0.782),
<PAD> (0.780)
  Large                   british (0.906),  day (0.893),  same (0.887),  team
(0.883),  first (0.881)

  Query: 'world'
  Model                   Neighbours

  Small                   waldrons (0.415),  1911 (0.410),  has (0.404),  such
(0.399),  form (0.395)
  Medium                  gods (0.749),  game (0.748),  of (0.746),  while
(0.738),  for (0.735)
  Large                   game (0.872),  british (0.853),  second (0.850),
early (0.833),  league (0.830)

  Query: 'war'
  Model                   Neighbours

  Small                   the (0.510),  were (0.503),  and (0.497),  was
(0.493),  rock (0.473)
  Medium                  between (0.643),  only (0.625),  may (0.624),  was
(0.615),  with (0.609)
  Large                   season (0.870),  new (0.866),  british (0.863),  first
(0.860),  all (0.849)

  Query: 'king'
  Model                   Neighbours

  Small                   (word not in vocabulary)
  Medium                  blackie (0.525),  during (0.522),  they (0.498),  to
(0.496),  has (0.488)
  Large                   project (0.827),  season (0.753),  team (0.753),  rock
(0.752),  building (0.750)
```

```
  Query: 'first'
  Model                    Neighbours

  Small                    once (0.488),  by (0.447),  unk (0.432),  that
(0.431),  an (0.422)
  Medium                   for (0.859),  with (0.833),  and (0.832),  is (0.831),
in (0.828)
  Large                    second (0.936),  season (0.917),  new (0.908),  team
(0.904),  best (0.903)
```

[ ]:
```python
import matplotlib.colors as mcolors

TOP_N = 150   # most-frequent words to visualise

def plot_pca_embeddings(embeddings_tensor, tokenizer_obj, top_n=TOP_N,␣
 ↪title_suffix=""):
    freq = tokenizer_obj.word_freq.copy()
    freq[tokenizer_obj.PAD_INDEX] = 0
    top_indices = np.argsort(freq)[::-1][:top_n]
    top_words   = [tokenizer_obj.index_to_word[i] for i in top_indices]
    top_freq    = freq[top_indices]

    vecs_np = embeddings_tensor.detach().numpy()[top_indices]
    pca     = PCA(n_components=2, random_state=42)
    reduced = pca.fit_transform(vecs_np)
    var_exp = pca.explained_variance_ratio_

    # Color by log-frequency quartile
    log_freq  = np.log1p(top_freq)
    norm      = mcolors.Normalize(vmin=log_freq.min(), vmax=log_freq.max())
    cmap      = plt.cm.viridis
    colors    = cmap(norm(log_freq))

    fig, ax = plt.subplots(figsize=(13, 9))
    sc = ax.scatter(
        reduced[:, 0], reduced[:, 1],
        c=log_freq, cmap="viridis", s=22, alpha=0.75, linewidths=0,
    )
    for i, word in enumerate(top_words):
        ax.annotate(
            word,
            (reduced[i, 0], reduced[i, 1]),
            fontsize=7,
            alpha=0.85,
            xytext=(2, 2),
            textcoords="offset points",
```

```
        )

    cbar = plt.colorbar(sc, ax=ax, pad=0.01)
    cbar.set_label("log(word frequency)", fontsize=9)

    ax.set_title(
        f"PCA of top-{top_n} word embeddings{title_suffix}\n"
        f"PC1 {var_exp[0]*100:.1f}% variance · PC2 {var_exp[1]*100:.1f}%␣
↪variance",
        fontsize=12, fontweight="bold",
    )
    ax.set_xlabel("Principal Component 1", fontsize=10)
    ax.set_ylabel("Principal Component 2", fontsize=10)
    ax.grid(True, alpha=0.15, linestyle=":")
    plt.tight_layout()
    plt.show()

plot_pca_embeddings(
    embedding_layer_large.weights,
    tokenizer_large,
    title_suffix=" - Large corpus",
)
```

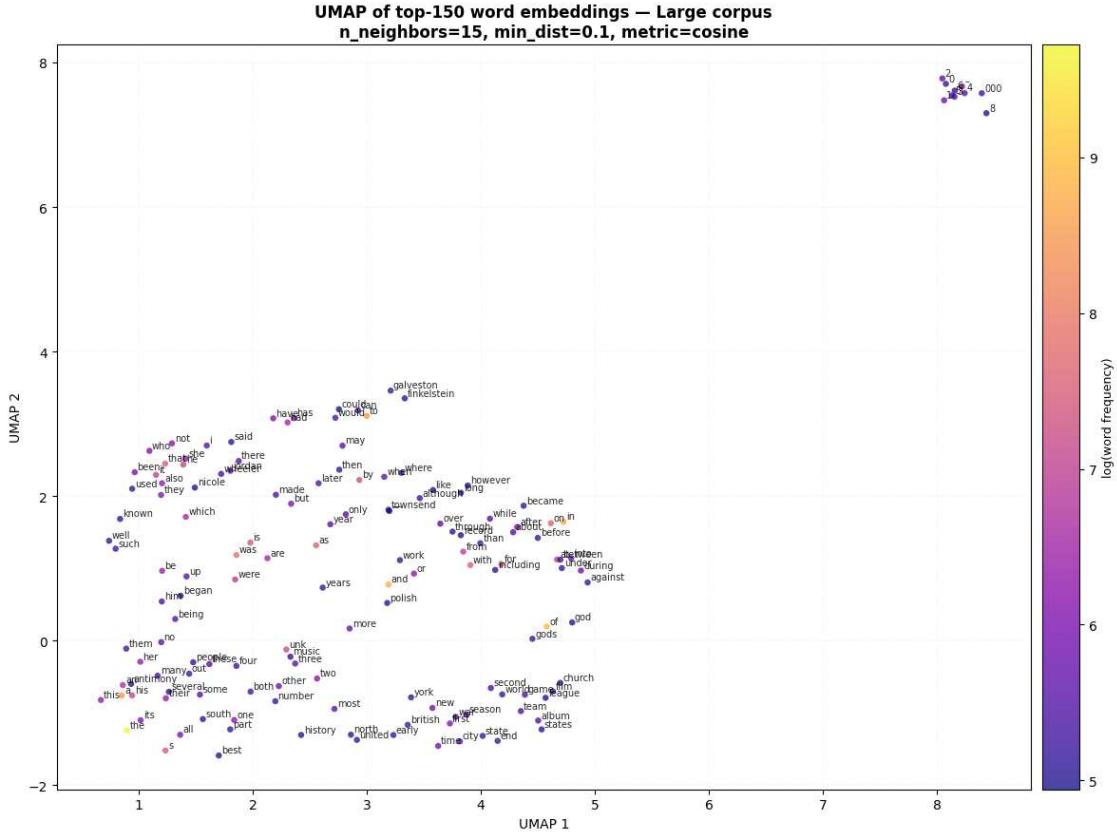


7.5.3 UMAP

[ ]:
```
def plot_umap_embeddings(embeddings_tensor, tokenizer_obj, top_n=150,␣
↪title_suffix="",
                         n_neighbors=15, min_dist=0.1):
    freq = tokenizer_obj.word_freq.copy()
    freq[tokenizer_obj.PAD_INDEX] = 0
    top_indices = np.argsort(freq)[::-1][:top_n]
    top_words   = [tokenizer_obj.index_to_word[i] for i in top_indices]
    top_freq    = freq[top_indices]

    vecs_np = embeddings_tensor.detach().numpy()[top_indices]

    reducer = umap.UMAP(
        n_components=2,
        n_neighbors=n_neighbors,
        min_dist=min_dist,
        random_state=42,
        metric="cosine",
    )
    reduced  = reducer.fit_transform(vecs_np)
    log_freq = np.log1p(top_freq)

    fig, ax = plt.subplots(figsize=(13, 9))
    sc = ax.scatter(
        reduced[:, 0], reduced[:, 1],
        c=log_freq, cmap="plasma", s=22, alpha=0.75, linewidths=0,
    )
    for i, word in enumerate(top_words):
        ax.annotate(
            word,
            (reduced[i, 0], reduced[i, 1]),
            fontsize=7,
            alpha=0.85,
            xytext=(2, 2),
            textcoords="offset points",
        )

    cbar = plt.colorbar(sc, ax=ax, pad=0.01)
    cbar.set_label("log(word frequency)", fontsize=9)

    ax.set_title(
        f"UMAP of top-{top_n} word embeddings{title_suffix}\n"
        f"n_neighbors={n_neighbors}, min_dist={min_dist}, metric=cosine",
```

```
        fontsize=12, fontweight="bold",
    )
    ax.set_xlabel("UMAP 1", fontsize=10)
    ax.set_ylabel("UMAP 2", fontsize=10)
    ax.grid(True, alpha=0.15, linestyle=":")
    plt.tight_layout()
    plt.show()

plot_umap_embeddings(
    embedding_layer_large.weights,
    tokenizer_large,
    title_suffix=" - Large corpus",
)
```

```
/home/enarde/jupyterlab-env/lib/python3.12/site-packages/umap/umap_.py:1952:
UserWarning: n_jobs value 1 overridden to 1 by setting random_state. Use no seed
for parallelism.
  warn(
```



[ ]:
```
# Summary comparison across all three trained models
_summary_configs = [
    ("Small",  tokenizer_small,  embedding_layer_small,  "ns_history_small"),
    ("Medium", tokenizer_medium, embedding_layer_medium, "ns_history_medium"),
    ("Large",  tokenizer_large,  embedding_layer_large,  "ns_history_large"),
]

rows = []
for label, tok, emb_layer, hist_var in _summary_configs:
    try:
        hist = eval(hist_var)
    except NameError:
        continue
    n_params = tok.VOCAB_SIZE * emb_layer.embed_dim   # input embeddings only
    rows.append({
        "Model":            label,
        "Vocab size":       tok.VOCAB_SIZE,
        "Embed params":     f"{n_params:,}",
        "Train pairs":      f"{int(tok.num_max_sentences * 0.8 * 20):,}",   #␣
 ↪rough estimate
        "Final train loss": f"{hist['train'][-1]:.4f}",
        "Final val loss":   f"{hist['val'][-1]:.4f}",
        "Epochs":           len(hist["train"]),
    })

df_summary = pd.DataFrame(rows).set_index("Model")
print("=" * 72)
print("  Model Comparison Summary")
print("=" * 72)
display(df_summary)

# PCA side-by-side for all three
_vis_configs = [
    (embedding_layer_small,  tokenizer_small,  "Small"),
    (embedding_layer_medium, tokenizer_medium, "Medium"),
    (embedding_layer_large,  tokenizer_large,  "Large"),
]

fig, axes = plt.subplots(3, 1, figsize=(10, 18))
for ax, (emb_layer, tok, label) in zip(axes, _vis_configs):
    freq = tok.word_freq.copy()
    freq[tok.PAD_INDEX] = 0
    top_n   = min(100, tok.VOCAB_SIZE - 1)
    top_idx = np.argsort(freq)[::-1][:top_n]
    words   = [tok.index_to_word[i] for i in top_idx]
    log_f   = np.log1p(freq[top_idx])
```

```
    vecs    = emb_layer.weights.detach().numpy()[top_idx]
    pca     = PCA(n_components=2, random_state=42)
    red     = pca.fit_transform(vecs)
    var     = pca.explained_variance_ratio_

    sc = ax.scatter(red[:, 0], red[:, 1], c=log_f, cmap="viridis",
                    s=18, alpha=0.75, linewidths=0)
    for i, w in enumerate(words):
        ax.annotate(w, (red[i, 0], red[i, 1]), fontsize=6, alpha=0.8,
                    xytext=(1, 1), textcoords="offset points")
    ax.set_title(
        f"{label}\nPC1 {var[0]*100:.1f}% · PC2 {var[1]*100:.1f}%",
        fontsize=10, fontweight="bold",
    )
    ax.set_xlabel("PC1", fontsize=9)
    ax.set_ylabel("PC2", fontsize=9)
    ax.grid(True, alpha=0.15, linestyle=":")
    plt.colorbar(sc, ax=ax, pad=0.01).set_label("log freq", fontsize=8)

fig.suptitle(
    "PCA Comparison - Word Embeddings Across Corpus Sizes (top-100 words)",
    fontsize=12, fontweight="bold", y=1.02,
)
plt.tight_layout()
plt.show()
```

[ ]:

```

```